

Spectral-front identities for periodic Navier–Stokes and an obstruction to uniform pointwise Osgood closure

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Abstract

For strong solutions of the unforced incompressible Navier–Stokes equations on $\mathbb{T}^3$ we prove that the spectral bandwidth $N_0^2 = \|\nabla u\|_2^2/\|u\|_2^2$ obeys the differential inequality $\frac{d}{dt} \log N_0^2 \leq KD/(2\nu)$, where $D = \|\nabla u\|_2^2$ and $K = \|\mathbb{P}(u \cdot \nabla u)\|_2^2/\|\nabla u\|_2^4$, with an explicit nonnegative defect; consequently if $\int_0^{T_{\max}} KD dt < \infty$ the solution is global, and in particular any finite-time breakdown forces the action to diverge. The action admits the exact representation $\int_0^{T'} KD dt = \int_0^{T'} \|\partial_t u\|_2^2/D dt + \nu^2 \int_0^{T'} \|\Delta u\|_2^2/D dt + \nu \log(D(T')/D(0))$; by AM–GM this yields the one-sided criterion $\int KD \leq 2 \int (\|\partial_t u\|_2^2/D + \nu^2 \|\Delta u\|_2^2/D)$ (the converse comparison fails, and the failure is one about the *infinite-time* integral $T' \rightarrow \infty$, not about any finite T'), and KD is dominated pointwise by both the Serrin $(\infty, 2)$ and the critical $\|\nabla u\|_{L^3}^2$ actions. On the obstruction side we construct an explicit family of divergence-free trigonometric fields with exact closed-form spectral laws, and prove for it the capacity lower bound $\|\mathbb{P}(u_N \cdot \nabla u_N)\|_2^2 \geq c_0 N^3$ with the sharp exponent, for every admissible cutoff profile χ and every nonzero seed vector v_0 ; the constants $c_0 = c_0(\chi, v_0)$ and $N_* = N_*(\chi, v_0)$ are *not* uniform in (χ, v_0) and are non-effective. It follows *unconditionally* that no Osgood-admissible pointwise field inequality $K \leq \Phi(\log N_0^2)$ can hold uniformly over all zero-mean divergence-free trigonometric fields: the dichotomy cannot be upgraded to unconditional regularity through that route. The capacity constant c_0 and the band threshold N_* are the only non-effective constants in the paper; every other constant is explicit. Proofs are elementary and, modulo classical local theory and the standard Fourier-analytic facts cited in the bibliography, complete and self-contained: the capacity bound is proved in full in Appendix B. No numerical statement enters any proof; all numerics are confined to a certificate appendix, and the machine-checkable certificates themselves remain in the repository. The claimed novelty is the exact packaging — the modal identities, their machine-checkable defect ledger, and the obstruction family with its capacity bound. The criterion is not claimed to be new in strength: it belongs to the family of dynamic-wavenumber criteria whose closest representative is that of Cheskidov–Shvydkoy [19], and the objects N_0 , N_1 and K are classical (Section 4). Internal review was performed by AI agents and is not human peer review.

1 Introduction

The regularity theory of the three-dimensional Navier–Stokes equations is governed by critical quantities whose a-priori control is unknown. We isolate a scalar action, $\int KD dt$, that (i)

*Machine-assisted research repository. All mathematics is contained in this paper; the exact-arithmetic certificates reported in the certificate appendix are reproducible from the repository scripts named there, and are not used in any proof. This draft claims no resolution of any Clay Millennium problem, in either direction.

bounds the growth of the spectral bandwidth through an exact modal identity with computable defects, (ii) is dominated by the classical critical actions, so its finiteness is a weaker hypothesis than theirs, and (iii) possesses an exact dynamic representation splitting it into a supercritical part (the $\dot{H}^1$ -bandwidth action) and a harmless logarithm. We then show, *unconditionally*, that the hypothesis cannot be verified by any pointwise Osgood-type field inequality: an explicit smoothly truncated coherent critical-spectrum family obeys an exact capacity lower bound with the sharp exponent, which forces any uniform majorant $\Phi(\log N_0^2)$ of K to be of exponential class. The pair of results freezes this lane: the dichotomy is at least as strong as the classical criteria it refines, and the pointwise-Osgood route to making it unconditional is closed.

We emphasise what is *not* claimed: no global regularity for all data, no blow-up, no Clay statement. Numerics never enter any proof; they appear only as reproducible exact-arithmetic certificates corroborating the proven statements.

Structure, and what is proved where. Section 3 gives the lemma chain behind Theorem 1 in compressed but complete form; each step is a few lines of elementary computation. Theorem 2, the one statement whose proof is long, is proved *in full* in Appendix B: cutoff class and exact family laws (§B.2–§B.3), the continuum profile V and its far field (§B.4–§B.5), the exact lattice-Riemann identity with an explicit rate (§B.6), the concentrated test field and the resulting N^3 duality lower bound (§B.7), the χ -independent non-degeneracy $\mathbb{P}(V \cdot \nabla V) \neq 0$ (§B.8), and the assembly (§B.9). Section 4 places the criterion among the known frequency-localised and dynamic-wavenumber criteria. Appendix A reports the numerical corroboration; the certificates themselves, being machine-checkable artefacts rather than mathematics, remain in the repository. Apart from the classical inputs listed after Theorem 1 and the standard Fourier-analytic facts isolated as Remarks 6–10, the paper is self-contained.

2 Setting and main results

Let $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$ with normalised measure, $u(x, t) = \sum_{k \in \mathbb{Z}^3 \setminus \{0\}} \hat{u}_k(t) e^{ik \cdot x}$ real, zero-mean, divergence-free ($k \cdot \hat{u}_k = 0$), solving $\partial_t u + \mathbb{P}(u \cdot \nabla u) = \nu \Delta u$ with $\nu > 0$ and $u_0 \in H_\sigma^m$, $m > 5/2$, $u_0 \neq 0$; let u be the maximal strong solution on $[0, T_{\max})$. Write $H_r = \sum_k |k|^{2r} |\hat{u}_k|^2$, $D = H_1$, $N_0^2 = H_1/H_0$, $N_1^2 = H_2/H_1$, $z = \log N_0^2$ (≥ 0), $\mathcal{N} = -\mathbb{P}(u \cdot \nabla u)$, and $K = \|\mathcal{N}\|_2^2/D^2$. The functional setting (Leray projection $\mathbb{P}$, Stokes operator, the spaces H_σ^m on the torus) is the standard one [4, 5], and the existence of a maximal strong solution with the continuation principle is classical [2, 1, 3, 9]. The ratios N_0 and N_1 are not new objects: $N_0 = \|\nabla u\|_2/\|u\|_2$ is the classical Taylor-microscale inverse length scale and N_1 the next rung of the same ladder [27, 28, 29, 30, 31, 32].

Theorem 1. *With the notation above:*

- (a) $\Lambda(t) = \log N_0^2(t) - \frac{1}{2\nu} \int_0^t KD \, ds$ is non-increasing on $[0, T_{\max})$; quantitatively $\frac{d}{dt} \log N_0^2 = \frac{2\text{Cov}}{N_0^2} - \frac{2\nu V}{N_0^2} \leq \frac{KD}{2\nu}$ with $\text{Cov} = T_1/H_0$, $V = \text{Var}_p(|k|^2)$, $p(k) = |\hat{u}_k|^2/H_0$, and an explicit nonnegative two-part defect.
- (b) Exactly one of: $\int_0^{T_{\max}} KD \, dt < \infty$, and then z is bounded, $u \in L^\infty(0, T_{\max}; H^1)$ and $T_{\max} = \infty$; or $\int_0^{T_{\max}} KD \, dt = \infty$. In particular $T_{\max} < \infty \Rightarrow \int_0^{T_{\max}} KD \, dt = \infty$; no claim is made that a global solution has finite action.

(c) For every $T' < T_{\max}$,

$$\int_0^{T'} KD dt = \int_0^{T'} \frac{\|\partial_t u\|_2^2}{D} dt + \nu^2 \int_0^{T'} N_1^2 dt + \nu \log \frac{D(T')}{D(0)},$$

and by AM–GM $\int_0^{T'} KD dt \leq 2 \int_0^{T'} (\|\partial_t u\|_2^2/D + \nu^2 N_1^2) dt$. The reverse comparison fails, and the counterexample is a statement about the infinite-time integral, i.e. about the limit $T' \rightarrow \infty$, not about a finite T' : for the global solution $u = e^{-\nu t}(0, 0, \cos x_1)$ one has $KD \equiv 0$, while the bandwidth action equals $\int_0^{T'} (\|\partial_t u\|_2^2/D + \nu^2 N_1^2) dt = 2\nu^2 T'$, which is finite for every finite T' and diverges only as $T' \rightarrow \infty$. For each finite T' all three terms of the identity are finite and the identity holds exactly, the deficit being carried by $\nu \log(D(T')/D(0)) = -2\nu^2 T'$. Thus no bound of the form $\int_0^{T'} (\|\partial_t u\|_2^2/D + \nu^2 N_1^2) dt \leq C \int_0^{T'} KD dt$ is available, and in particular finiteness of $\int_0^\infty KD dt$ does not imply finiteness of the infinite-time bandwidth action.

(d) Pointwise, $KD \leq \|u\|_{L^\infty}^2$ and $KD \leq C_S^2 \|\nabla u\|_{L^3}^2$.

(e) If moreover $KD \leq \Phi(z)D + R$ a.e. with $\Phi > 0$ nondecreasing, $\int^\infty ds/\Phi = \infty$, $R \geq 0$, $\int_0^{T_{\max}} R dt < \infty$, then the first alternative of (b) holds.

The proof (Section 3) is a chain of eight elementary lemmas; the only external inputs are the classical local theory **(F1)** [2, 1, 3, 9], the subcritical Ladyzhenskaya–Prodi–Serrin class $L^\infty L^6$ **(F2)** [6, 7, 8, 9], and the mean-zero Sobolev embedding $H^1 \hookrightarrow L^6$ **(F3)** [11]. We do not use the endpoint $L^\infty L^3$ theory of [10].

On the obstruction side, we use a *smoothly* truncated coherent family.

Definition 1 (admissible cutoff). $\chi \in C^4([0, \infty); [0, 1])$ with $\chi \equiv 1$ on $[0, \frac{1}{2}]$ and $\text{supp } \chi \subset [0, 1]$; no monotonicity is assumed. Write $c_\chi = 1 + \|\chi'\|_\infty$.

Remark 1 (what the regularity of χ is for). Only χ' is used quantitatively, through the single constant c_χ in the Riemann-sum rate behind Theorem 2 (Theorem 7); $\chi \in C^1$ — indeed χ merely Lipschitz — suffices for Theorem 2 and hence for Theorem 3. Derivatives of order 2, 3, 4 occur once only, in the far-field expansion $V = \pi^2 W + O(|y|^{-4})$ (Theorem 6), which no proof consumes. The non-degeneracy $\mathbb{P}(V \cdot \nabla V) \not\equiv 0$ uses no derivative of χ whatever — only $0 \leq \chi \leq 1$ and $\chi \equiv 1$ on $[0, \frac{1}{2}]$. The smoothness of V itself comes from the compact support of F , not from that of χ ; no step requires $\chi \in C^\infty$. The class is stated as C^4 , rather than as the load-bearing minimum C^1 , so that every statement of Appendix B holds verbatim and the exact-rational certificate cutoff of Appendix A is admissible.

Fix $v_0 \in \mathbb{R}^3 \setminus \{0\}$ (no integrality is needed) and an admissible χ , and let

$$\hat{u}_N(k) = \chi\left(\frac{|k|}{N}\right) \frac{P_k v_0}{|k|^2} \quad (k \neq 0), \quad \hat{u}_N(0) = 0, \quad P_k = I - \frac{k \otimes k}{|k|^2}, \quad (1)$$

a finite trigonometric polynomial banded at $|k| \leq N$.

Lemma 1 (exact family laws). u_N is real, zero-mean and divergence-free, and with $S_N^\chi = \sum_{k \neq 0} \chi(|k|/N)^2 |k|^{-2}$, $T_N^\chi = \sum_{k \neq 0} \chi(|k|/N)^2 |k|^{-4}$, $\Sigma_N^\chi = \sum_{k \neq 0} \chi(|k|/N) |k|^{-2}$: $H_0 = \frac{2}{3} \|v_0\|^2 T_N^\chi$,

$H_1 = \frac{2}{3}\|v_0\|^2 S_N^\chi$, $N_0^2 = S_N^\chi/T_N^\chi$, $u_N(0) = \frac{2}{3}\Sigma_N^\chi v_0$, and $\|u_N\|_\infty^2/\|\nabla u_N\|_2^2 \geq 2(\Sigma_N^\chi)^2/(3S_N^\chi)$. Moreover, for $N \geq 32$ and every admissible χ ,

$$\frac{N}{348} \leq S_N^\chi \leq 128N, \quad 6 \leq T_N^\chi \leq 128, \quad \frac{N}{44544} \leq N_0^2 \leq \frac{64}{3}N,$$

all crude explicit constants, proven; so $N_0^2 \asymp N$.

The band-symmetry identity behind Lemma 1 is needed for a *general* radial weight f with $\sum |k|^2 |f| < \infty$ — no positivity and no sharp ball — since χ is not assumed monotone.

Theorem 2 (capacity lower bound; sharp exponent). *For every $v_0 \in \mathbb{R}^3 \setminus \{0\}$ and every admissible χ there exist $c_0 = c_0(\chi, v_0) > 0$ and $N_* = N_*(\chi, v_0)$ with*

$$\|\mathbb{P}(u_N \cdot \nabla u_N)\|_2^2 \geq c_0 N^3 \quad \text{for all } N \geq N_*.$$

The exponent 3 is sharp, since (d) with Lemma 1 gives $\|\mathbb{P}(u_N \cdot \nabla u_N)\|_2^2 \leq \|u_N\|_\infty^2 H_1 \asymp N^3$. The quantifiers matter: the bound holds for every admissible χ and every $v_0 \neq 0$, but c_0 and N_* depend on the pair (χ, v_0) and are **not** uniform in it; they are also non-effective (Remark 3). What is uniform in (χ, v_0) is the underlying non-degeneracy $\mathbb{P}(V \cdot \nabla V) \neq 0$, which is what makes the quantifier “for every” legitimate.

Theorem 2 replaces the capacity *hypothesis* of earlier versions of this paper. Its complete proof is given in Appendix B of this paper; Section 3 gives a summary only. (The machine-checkable exact-arithmetic certificates that corroborate it remain in the repository, and are reported in Appendix A.)

Theorem 3 (no Osgood-admissible uniform majorant). *If Φ is nondecreasing and $K(u) \leq \Phi(\log N_0^2(u))$ for every zero-mean divergence-free real trigonometric field u , then $\Phi(s) \geq ce^s$ for all large s ; in particular $\int^\infty ds/\Phi < \infty$, so clause (e) of Theorem 1 cannot be activated by any uniform pointwise field inequality.*

Remark 2 (what is open, and no longer used). *The sharply truncated family $\hat{u}_N(k) = P_k v_0/|k|^2$ on $1 \leq |k| \leq N$ satisfies the same exact laws with S_N, T_N in place of S_N^χ, T_N^χ and saturates the Bernstein ratio two-sidedly, $\frac{2}{3}S_N \leq \|u_N\|_\infty^2/\|\nabla u_N\|_2^2 \leq S_N$. The corresponding capacity bound for that family — this paper’s former Hypothesis (L^*) — remains **open**. It is now **unused**: Theorem 3 quantifies over all real zero-mean divergence-free trigonometric fields, so the smooth family alone suffices. The honest statement is “the obstruction is unconditional”, not “ (L^*) is proven”. A proven negative result constrains the sharp family further: no constant-vector sweeping split, in any Hölder pairing and any constant pairing direction, can exceed $\|\mathbb{P}(u_N \cdot \nabla u_N)\|_2^2 \gtrsim N$ — two powers short — which is why a concentrated test field, not a constant sweep, is the right instrument.*

Remark 3 (effectivity). *The constants c_0 and N_* of Theorem 2, and hence c of Theorem 3, depend on (χ, v_0) and are **non-effective** (see Remark 13): the test field driving the capacity bound is produced from the density of $C_{c,\sigma}^\infty(\mathbb{R}^3)$ in $L_\sigma^2(\mathbb{R}^3)$, a pure existence argument. Every other constant in this paper is explicit. An effective version would require an explicitly constructed divergence-free Ψ with $\int (V \cdot \nabla V) \cdot \Psi \neq 0$ together with a quantitative lower bound on that pairing; non-vanishing is certified only on $\{0 < |\zeta| < \pi^2/2048\}$ intersected with two cone conditions, so such a Ψ has spatial radius $R \gtrsim 10^3$ and forces $N_* \gtrsim 10^4$.*

3 Proofs

The lemma chain L0–L11 is given here in compressed form; the same chain, written out at length, is in the repository file `complete_proof.md`. The one substantial proof of the paper, that of Theorem 2, is not compressed: it is given complete in Appendix B.

L0a (regularity of moments). H_0, H_1, H_2 are continuous; $H_0, H_1 \in C^1$ with $\dot{H}_0 = 2\langle u, \partial_t u \rangle$, $\dot{H}_1 = 2\langle -\Delta u, \partial_t u \rangle$ (difference quotients in $C^1 H^{m-2}$ paired against $C(L^2)$); $\partial_t u = \mathcal{N} - \nu(-\Delta)u$ in L^2 . Positivity of H_0 on all of $[0, T_{\max})$ is *not* elementary and is proven as L0b *after* L7, by bootstrap: if t_1 is the first vanishing time, then on $[0, t_1)$ Lemmas 1–4 and 7 apply, $KD \leq \|u\|_\infty^2 \leq C_m M^2$ with $M = \sup_{[0, t_1]} \|u\|_{H^m}$, so N_0^2 is bounded there, $\int_0^{t_1} N_0^2 < \infty$, and the integrated energy identity gives $H_0(t_1) > 0$, a contradiction.

L1 (ledger; energy neutrality). $\frac{1}{2}\dot{H}_r = T_r - \nu H_{r+1}$ ($r = 0, 1$) with $T_r = \sum |k|^{2r} \text{Re}\langle \hat{u}_k, \hat{\mathcal{N}}_k \rangle$; $T_0 = -\langle u, (u \cdot \nabla)u \rangle = 0$; hence $\int_0^{T_{\max}} D dt \leq H_0(0)/(2\nu)$.

L2 (identity). $\frac{d}{dt} \log N_0^2 = 2\text{Cov}/\mu - 2\nu V/\mu$ with $\mu = N_0^2$: subtract the logarithmic derivatives and use $H_0 H_2 - H_1^2 = H_0^2 V$.

L3 (closable covariance bound). $|\text{Cov}| \leq \sqrt{V\|\mathcal{N}\|_2^2/H_0}$: modal Cauchy–Schwarz $|\text{Re}\langle \hat{u}_k, \hat{\mathcal{N}}_k \rangle| \leq |\hat{u}_k| |\hat{\mathcal{N}}_k|$, then vector Cauchy–Schwarz.

L4 (square completion; (a)). $\sup_{V \geq 0} [\alpha\sqrt{V} - \nu V] = \alpha^2/(4\nu)$ with $\alpha = \sqrt{\|\mathcal{N}\|_2^2/H_0}$ gives $\frac{d}{dt} \log N_0^2 \leq \|\mathcal{N}\|_2^2/(2\nu H_1) = KD/(2\nu)$; the defect is $\Gamma^{\text{CS}} + \Gamma^{\text{SC}} \geq 0$ as displayed in the repository file, with rational total.

L5 (dichotomy; (b)). $\int KD < \infty \Rightarrow z \leq z_* \Rightarrow H_1 \leq e^{z_*} H_0(0) \Rightarrow u \in L^\infty L^6$ (mean-zero Sobolev), a subcritical Serrin class $\Rightarrow$ regularity and extension.

L6 (action representation; (c)). Expand $\|\mathcal{N}\|_2^2 = \|\partial_t u + \nu(-\Delta)u\|_2^2$, use $2\nu\langle \partial_t u, -\Delta u \rangle = \nu\dot{D}$, divide by $D > 0$, integrate; $\int \dot{D}/D = \log(D(T')/D(0))$.

L7 (dominations; (d)). $\|\mathcal{N}\|_2 \leq \|u\|_\infty \|\nabla u\|_2$ and $\|\mathcal{N}\|_2 \leq \|u\|_{L^6} \|\nabla u\|_{L^3} \leq C_S \|\nabla u\|_2 \|\nabla u\|_{L^3}$. Here C_S is any admissible constant for the mean-zero embedding $H^1(\mathbb{T}^3) \hookrightarrow L^6(\mathbb{T}^3)$ [11]; no sharp value is used or claimed. (On $\mathbb{R}^3$ the sharp constant is Talenti’s [12]; we make no assertion that it transfers to the mean-zero torus.)

L8 (Osgood closure; (e)). $\Omega(z) = \int_0^z ds/\Phi$; $\frac{d}{dt}\Omega(z) \leq D/(2\nu) + R/(2\nu\Phi(0))$, both integrable; z bounded; apply L5.

L9–L11 and Theorem 3. Band symmetry $\sum_{k \neq 0} k_i k_j f = \delta_{ij} \frac{1}{3} \sum |k|^2 f$ for any radial f with $\sum |k|^2 |f| < \infty$ (sign flips kill $i \neq j$; permutations equalise the diagonal; no positivity is used); the family laws of Lemma 1 follow by applying it with the weights $\chi(|k|/N)^2 |k|^{-6}$, $\chi(|k|/N)^2 |k|^{-4}$ and $\chi(|k|/N) |k|^{-4}$ to $|P_k v_0|^2 = \|v_0\|^2 - (k \cdot v_0)^2/|k|^2$ and to $u_N(0) = \sum \hat{u}_N(k)$. The lattice

bounds follow from the dyadic shell count $\#\{2^j \leq |k| < 2^{j+1}\} < 64 \cdot 8^j$ (upper) and an all-positive sub-box count in $N/2 < |k| \leq N$ (lower), using $\chi \equiv 1$ on $[0, \frac{1}{2}]$ for $S_N^\chi \geq S_{\lfloor N/2 \rfloor}$ and the six unit vectors for $T_N^\chi \geq 6$; the two-sided bound gives 2^q -adic gaps $c_- \leq s_{2^q N} - s_N \leq c_+$ with $q = 20$, $c_- = \log(1048576/950272) = 0.0985\dots$, $c_+ = \log(2^{20} \cdot 950272) = 27.63\dots$, elementary and N -uniform. By Theorem 2 and $H_1 \leq \frac{256}{3}\|v_0\|^2 N$, $K(u_{N_j}) \geq c_0'' e^{s_{N_j}}$ along $N_j = 2^{20j} N_\sharp$; monotonicity of Φ interpolates across the bounded gaps, so $\Phi(s) \geq c_0'' e^{-c_+} e^s$ for all large s and $\int^\infty ds/\Phi < \infty$.

Theorem 2 (summary; complete proof in Appendix B). Because $P_k = P_{k/N}$, the family obeys the exact scaling $\hat{u}_N(k) = N^{-2}F(k/N)$ with $F(\xi) = \chi(|\xi|)P_\xi v_0/|\xi|^2$, so with $V = \tilde{F}$ (inverse transform on $\mathbb{R}^3$) the lattice sums defining $u_N(y/N)$ and $\nabla u_N(y/N)$ are *literally* Riemann sums of spacing $1/N$: $u_N(y/N) = N(V(y) + E_N(y))$, $\nabla u_N(y/N) = N^2(\nabla V(y) + E'_N(y))$ identically, with $\sup_{|y| \leq R}(|E_N| + |E'_N|) = O(N^{-1} \log N)$ explicitly (the integrand is compactly supported with a single integrable algebraic singularity; the gradient half needs and receives its own, log-free, argument). The naive periodisation route — writing u_N as an absolutely convergent sum $N \sum_m V(N(x + 2\pi m))$ over lattice translates — is *unavailable*: V decays only like $|y|^{-1}$, with a sign-definite spherical mean in the v_0 direction, so the series of translates diverges and the naive absolutely convergent periodisation argument cannot be run. (We make no claim about summation methods in general; the point is simply that this argument is not available, and none is needed — an exact lattice-Riemann-sum identity replaces it. See Remark 8.) Pairing against the concentrated divergence-free test field $\psi_N(x) = N^{3/2}\Psi(Nx)$, whose $L^2(\mathbb{T}^3)$ norm is N -independent, and using self-adjointness of $\mathbb{P}$ gives $\|\mathbb{P}(u_N \cdot \nabla u_N)\|_2^2 \geq (2\pi)^{-3} N^3 |I_\Psi + \mathcal{E}_N|^2 / \|\Psi\|_{L^2(\mathbb{R}^3)}^2$ with $I_\Psi = \int (V \cdot \nabla V) \cdot \Psi$ and $\mathcal{E}_N = O(\log N/N)$. It remains to find one Ψ with $I_\Psi \neq 0$, i.e. to show $\mathbb{P}(V \cdot \nabla V) \neq 0$. This holds for *every* admissible χ and *every* v_0 , because the leading small- ζ singularity of $F * F$ is the χ -independent convolution τ built from the Oseen tensor [13, 14] — evaluated by the classical Fourier transforms of homogeneous distributions [16, 15, 17] — for which, with $v_0 = \|v_0\|e_3$,

$$\zeta \times (\tau(\zeta)\zeta) = \frac{3\pi^3}{8} \|v_0\|^2 \frac{\zeta_3}{|\zeta|} (\zeta \times e_3) \neq 0$$

off the two degenerate directions — the Fourier form of the elementary fact that the Oseen field is not a stationary Euler flow, since $\nabla \times (W \times \nabla \times W) = 12y_3(e_3 \times y)/|y|^6$. A crude perturbation bound $|T - \tau| \leq 64\pi\|v_0\|^2$ on $|\zeta| \leq \frac{1}{8}$ transfers this to the cutoff family on $|\zeta| < \pi^2/2048$. $\square$

4 Position among known criteria; limitations

By (d), the finiteness of $\int KD$ is implied by the classical Serrin $(\infty, 2)$ action and by $\int \|\nabla u\|_{L^3}^2 dt$; the criterion (b) is therefore at least as strong as both (we do not claim strictly stronger). By (c) and its AM–GM corollary, the criterion is implied by the a-priori control of $\int (\|\partial_t u\|_2^2/D + \nu^2 N_1^2) dt$ — the half-derivative-supercritical $\dot{H}^1$ -bandwidth action — while the exact identity shows the two differ only by $\nu \log D$, which can diverge to $-\infty$ on decaying solutions; this makes precise both why the criterion is strong and why its unconditional verification is out of reach of the present methods. The comparison is genuinely one-sided, and the failure of the reverse comparison is an *infinite-time* phenomenon: as recorded in Theorem 1(c), for $u = e^{-\nu t}(0, 0, \cos x_1)$ the bandwidth action is $2\nu^2 T'$, finite for every finite T' and divergent only in

the limit $T' \rightarrow \infty$, while $\int KD \equiv 0$ throughout. No finite- T' counterexample is claimed or needed.

Comparison with the frequency-localised and dynamic-wavenumber criteria. The criterion $\int_0^{T_{\max}} KD dt < \infty$ belongs to the family of regularity criteria stated in terms of a dynamically defined wavenumber rather than a norm, whose closest and strongest representative is the dissipation-wavenumber criterion of Cheskidov and Shvydkoy [19]: they define $\Lambda(t)$ through Littlewood–Paley localisation and prove regularity under $\Lambda \in L^{5/2}(0, T)$, a hypothesis they show to be weaker than *every* Ladyzhenskaya–Prodi–Serrin condition, whereas our (d) only establishes that $\int KD dt < \infty$ is *implied* by the Serrin $(\infty, 2)$ and $\int \|\nabla u\|_{L^3}^2$ actions; we therefore do not claim our criterion is new in strength, and it may well be strictly stronger — i.e. a more demanding hypothesis — than [19].

What is genuinely different here is the mechanism rather than the class: K and N_0 are L^2 -quadratic Fourier ratios, so the criterion is an exact differential identity for $\frac{d}{dt} \log N_0^2$ with a computable nonnegative defect, whereas the frequency-localised criteria of [18, 19, 20, 21] are dyadic-block estimates in Besov-type norms and carry unquantified harmonic-analysis constants. Conversely the ingredients we use are not new: N_0 and N_1 are the classical Taylor-microscale and next-rung inverse length scales of [27, 29, 30, 31, 32], and the smallness of $\|\mathbb{P}(u \cdot \nabla u)\|_2$ relative to $\|\nabla u\|_2^2$ that K measures is the “depletion of nonlinearity” studied in [33]. The supercriticality of $\int N_1^2$ in (c) is the familiar one: N_1^{-1} is a dissipative small scale, controlled a priori only through Gevrey/analyticity-radius estimates [34, 35].

Overlap is therefore concrete, and we state it as such rather than hedging: any solution satisfying a Serrin condition, or the $\dot{B}_{\infty, \infty}^0$ /BMO logarithmic criteria of [22, 23] in the range where those imply $u \in L^\infty L^6$, also satisfies the finite-action hypothesis of Theorem 1(b), so the criterion is *not* independent of the classical ones. We are not aware of the specific quantity $\int \|\mathbb{P}(u \cdot \nabla u)\|_2^2 / \|\nabla u\|_2^2 dt$, nor of the monotonicity of $\log N_0^2 - \frac{1}{2\nu} \int KD$, appearing in the literature; that packaging — not the class of criterion — is what we claim, and we claim it as “not aware of”, not as priority. Finally, Theorem 3 does not contradict the logarithmic closures of [23, 24]: those bound one *norm* of a solution by a logarithm of a higher norm along the flow, whereas we exclude a *pointwise field inequality* $K(u) \leq \Phi(\log N_0^2(u))$ holding uniformly over all zero-mean divergence-free trigonometric fields, which is a strictly stronger demand and is what clause (e) would need. Results at the critical scale such as [25, 26] are context rather than comparison: they concern well- and ill-posedness, not criteria for a given strong solution.

Limitations. Theorem 3 closes the pointwise-Osgood route unconditionally. Its one limitation is effectivity (Remark 3, Remark 13): the capacity constant c_0 and the threshold N_* are not produced, only shown to exist, and they depend on (χ, v_0) . All statements are elementary; the contribution is the exact packaging, the obstruction family with its capacity bound, and the machine-verifiable defect ledger accompanying (a).

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A Certificate appendix (numerics, separated from all proofs)

No number below enters any proof. Both Theorem 1 and Theorem 3 are unconditional, so — unlike in earlier versions — these values no longer carry a hypothesis; they are corroboration.

Smooth family (the family used by Theorem 3). Reproducible from the repository (experiments/run_smooth_family_capacity.py; JSON with SHA-256 sidecar at outputs/1star/smooth_family). Seed $v_0 = (1, 2, 3)$; weight $\chi(r) = 1 - S(\text{clamp}((r^2 - \frac{1}{4})/\frac{3}{4}, 0, 1))$ with S the degree-9 smoothstep $126s^5 - 420s^6 + 540s^7 - 315s^8 + 70s^9$, which is a polynomial in r^2 — hence $\chi(|k|/N) \in \mathbb{Q}$ at every lattice point and the whole family is exactly rational. This χ lies in $C^4 \setminus C^5$, with $\chi \equiv 1$ on $[0, \frac{1}{2}]$, $\text{supp } \chi = [0, 1]$, $0 \leq \chi \leq 1$ and $\|\chi'\|_\infty = 5.2167\dots$ (so $c_\chi = 6.2168\dots$): it is therefore *admissible* in the sense of Definition 1, and the exact lanes below certify the family to which Theorem 2 literally applies. (See Remark 1: the proof never uses more than χ' quantitatively.) Degree- $(2p + 1)$ smoothsteps give rational $\chi \in C^p$ for every p , should a stronger class ever be wanted.

Exact-Fraction verification of the family laws at $N = 4, 6, 8, 10, 12$: every residual of H_0 , H_1 and $u_N(0)$ is the rational 0, with $u_N(0)$ exactly parallel to v_0 . Exact capacity by full $O(|B_N|^2)$ contraction over all ordered pairs of active band points:

N	4	6	8	10	12	16
$\ \mathbb{P}(u_N \cdot \nabla u_N)\ _2^2/N^3$	678.28	1027.31	1241.51	1384.36	1485.78	1619.65
K	0.6166	1.1359	1.6705	2.2077	2.7464	3.8254
K/N_0^2	0.2662	0.3287	0.3643	0.3861	0.4008	0.4195

A dealiased FFT continuation (grid $4N + 2$, whole product spectrum represented) extends this to $N = 48$: $\|\mathbb{P}(u_N \cdot \nabla u_N)\|_2^2/N^3$ rises $1241.5 \rightarrow 1911.3$ and K/N_0^2 rises $0.3643 \rightarrow 0.4574$ over $N = 8 \rightarrow 48$, both monotonically. Exact versus FFT at the two shared bands agrees to double-precision roundoff: relative difference 5.5×10^{-16} ($N = 8$) and 7.0×10^{-16} ($N = 16$) in the norm, 5.3×10^{-16} and 8.1×10^{-16} in K . Two-point Richardson diagnostics (not proofs): $\|\mathbb{P}(u_N \cdot \nabla u_N)\|_2^2/N^3 \rightarrow 2064.3$, $K/N_0^2 \rightarrow 0.4764$, $H_1/N \rightarrow 87.51$, $N_0^2/N \rightarrow 0.5671$ — positive limits, i.e. the sharp exponent, consistent with Theorem 2. The continuum non-degeneracy $\mathbb{P}(V \cdot \nabla V) \neq 0$ is corroborated three independent ways (exact rational lattice modes; float 3-D quadrature of the continuum convolution; an axisymmetric radial reduction giving $\text{curl}(V \cdot \nabla V) = Z(r) \sin \theta \cos \theta e_\phi$ with $Z \rightarrow -12\pi^4 \|v_0\|^2/r^4$), agreeing to 10^{-4} or better.

Sharp family (corroboration only; family no longer used). These are *sharp*-family numbers, supporting the open Hypothesis (L^*) of Remark 2, which no statement in this paper uses. Reproducible from `experiments/run_coherent_family_certificates.py` and `run_osgood_gate.py` (JSON with SHA-256 sidecars under `outputs/verification_sprint_v1/osgood_gate/`): exact-Fraction family laws at $N = 4, 6, 8$ (residual 0); exact $K(u_4) = 0.78841070\dots$ (full fraction stored), $K(u_6) = 1.43447181\dots$, $K(u_8) = 2.03724301\dots$; float pipeline agreement 0.0, 4.6×10^{-16} , 1.3×10^{-15} ; $Q(v_0) > 0$ exact at $N = 4, 6, 8$ with float continuation $Q/N^2 = 850.7 \rightarrow 1029.5$ over $N = 8 \rightarrow 32$; captured fraction of $\|\mathbb{P}(u \cdot \nabla u)\|_2$ by the sweeping pairing: 0.921/0.917/0.915 (exact, $N = 4, 6, 8$) and 0.915 $\rightarrow$ 0.912 (float, $N = 8 \rightarrow 32$); $K/N_0^2 = 0.259 \rightarrow 0.396$ ($N = 4 \rightarrow 32$); Leray retention 0.780 $\rightarrow$ 0.867.

B Complete proof of the capacity bound (Theorem 2)

This appendix proves Theorem 2 in full. Nothing below is quoted from elsewhere except four classical facts, each isolated and labelled as such (Remarks 6, 7, 9 and 10); no numerical statement is used anywhere.

The chain is: the family u_N has an exact scaling $\hat{u}_N(k) = N^{-2}F(k/N)$ (§B.4), so that the lattice sums reconstructing u_N near the origin are *literally* Riemann sums, of spacing $1/N$, for the inverse Fourier transform V of F (§B.6); pairing $\mathbb{P}(u_N \cdot \nabla u_N)$ against a test field concentrated at scale $1/N$ therefore converts, with an explicit error, into the fixed continuum pairing $\int_{\mathbb{R}^3} (V \cdot \nabla V) \cdot \Psi$ times $N^{3/2}$ (§B.7); and that continuum pairing is nonzero for some Ψ because $\mathbb{P}_{\mathbb{R}^3}(V \cdot \nabla V) \neq 0$, which is the Fourier form of the elementary fact that the Oseen field is not a stationary Euler flow (§B.8). The bookkeeping is assembled in §B.9.

B.1 Conventions

On $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$ we use the normalised measure $d\mu = (2\pi)^{-3}dx$, so that for $u : \mathbb{T}^3 \rightarrow \mathbb{R}^3$

$$u(x) = \sum_{k \in \mathbb{Z}^3} \hat{u}(k) e^{ik \cdot x}, \quad \hat{u}(k) = \int_{\mathbb{T}^3} u(x) e^{-ik \cdot x} d\mu(x),$$

$$\langle f, g \rangle = \int_{\mathbb{T}^3} f \cdot \bar{g} d\mu = \sum_{k \in \mathbb{Z}^3} \hat{f}(k) \cdot \overline{\hat{g}(k)}, \quad H_r(u) = \sum_{k \in \mathbb{Z}^3} |k|^{2r} |\hat{u}(k)|^2.$$

On $\mathbb{R}^3$ we use the forward transform and the inverse-transform bracket

$$\mathcal{F}f(\zeta) = \int_{\mathbb{R}^3} f(y) e^{-i\zeta \cdot y} dy, \quad \tilde{G}(y) = \int_{\mathbb{R}^3} G(\xi) e^{i\xi \cdot y} d\xi,$$

so that $\mathcal{F}\tilde{G} = (2\pi)^3 G$ and $\|\tilde{G}\|_{L^2(\mathbb{R}^3)}^2 = (2\pi)^3 \|G\|_{L^2(\mathbb{R}^3)}^2$.

For $\xi \in \mathbb{R}^3 \setminus \{0\}$ put $P_\xi = I - \frac{\xi \otimes \xi}{|\xi|^2}$. Thus P_ξ is the orthogonal projection onto $\xi^\perp$: it is symmetric, idempotent, $P_\xi \xi = 0$, $\|P_\xi\| \leq 1$, $P_{-\xi} = P_\xi$, it is homogeneous of degree 0 and smooth on $\mathbb{R}^3 \setminus \{0\}$, and consequently $P_k = P_{k/N}$ for every $k \neq 0$ and every $N > 0$. The Leray projection $\mathbb{P}$ acts on $\mathbb{T}^3$ by $\widehat{\mathbb{P}f}(k) = P_k \hat{f}(k)$ for $k \neq 0$, leaving the mean untouched, and on $\mathbb{R}^3$ by the same symbol P_ζ . On both $L^2(\mathbb{T}^3)$ and $L^2(\mathbb{R}^3)$, $\mathbb{P}$ is an orthogonal projection, hence self-adjoint.

Throughout, $\|v_0\|$ is the Euclidean norm of $v_0 \in \mathbb{R}^3$, and e_1, e_2, e_3 is the standard basis. Repeated Latin indices are summed. One notational warning: in this appendix V always denotes

the continuum profile of Definition 4; the unrelated use of V for the modal variance $\text{Var}_p(|k|^2)$ in Theorem 1(a) does not occur here.

B.2 The cutoff class, the family, and its exact laws

We restate the definitions of the main text for completeness.

Definition 2 (admissible cutoff; restatement of Definition 1). $\chi \in C^4([0, \infty); [0, 1])$ with $\chi \equiv 1$ on $[0, \frac{1}{2}]$ and $\text{supp } \chi \subset [0, 1]$. No monotonicity is assumed — it is used nowhere. Write $c_\chi := 1 + \|\chi'\|_{L^\infty} (\geq 1)$.

Such χ exist in abundance. The standard mollifier construction gives a C^∞ example, $\chi(t) = \Theta(2t - 1)$ with $\Theta(s) = 1$ for $s \leq 0$, $\Theta(s) = 0$ for $s \geq 1$ and $\Theta(s) = \frac{e^{-1/(1-s)}}{e^{-1/s} + e^{-1/(1-s)}}$ on $(0, 1)$; the exactly rational cutoff of Appendix A, built from the degree-9 smoothstep, lies in $C^4 \setminus C^5$ and is likewise admissible.

Remark 4 (what smoothness is actually used). *Definition 2 is still stronger than the proof requires. The only place any derivative of χ is used load-bearingly is Theorem 7, and it enters solely through $c_\chi = 1 + \|\chi'\|_\infty$; Sections B.7 and B.8 use only $0 \leq \chi \leq 1$, $\chi \equiv 1$ on $[0, \frac{1}{2}]$ and $\text{supp } \chi \subset [0, 1]$, and Sections B.2, B.3, B.4 use no derivative of χ at all. Every statement of this appendix on which Theorem 11 depends therefore holds verbatim for $\chi \in C^1([0, \infty); [0, 1])$ — indeed for Lipschitz χ — with those support properties. The four derivatives are demanded only so that Theorem 6 holds as stated with $M = 4$; that theorem is consumed only by the inert Remark 8, and nothing in the proof of Theorem 11 uses it. In particular $V \in C^\infty$ (Theorem 5(3)) and the Paley–Wiener statement there are bought by the compact support of F , not by smoothness of χ , and are unaffected.*

Definition 3. Fix $v_0 \in \mathbb{R}^3 \setminus \{0\}$, an admissible χ , and an integer $N \geq 2$. Define $u_N = u_N^{\chi, v_0}$ on $\mathbb{T}^3$ by

$$\hat{u}_N(k) = \chi\left(\frac{|k|}{N}\right) \frac{P_k v_0}{|k|^2} \quad (k \neq 0), \quad \hat{u}_N(0) = 0. \quad (2)$$

Since $\hat{u}_N(k) = 0$ for $|k| > N$, u_N is a finite trigonometric polynomial banded at $|k| \leq N$; its coefficients are real vectors and no integrality of v_0 is required.

Lemma 2 (basic structure). u_N is real, zero-mean, exactly divergence-free, C^∞ , and $u_N \neq 0$.

Proof. Zero mean is $\hat{u}_N(0) = 0$. Reality: the coefficients are real vectors and $P_{-k} = P_k$, so $\hat{u}_N(-k) = \hat{u}_N(k) = \overline{\hat{u}_N(k)}$, which is exactly the reality condition $\hat{u}(-k) = \overline{\hat{u}(k)}$. Divergence-free: $\widehat{\nabla \cdot u_N}(k) = ik \cdot \hat{u}_N(k)$ and $k \cdot P_k v_0 = 0$ for every $k \neq 0$. Smoothness: a finite sum of exponentials. Finally, among e_1, e_2, e_3 at least two are not parallel to v_0 ; pick such a k with $|k| = 1$. Then $\chi(1/N) = 1$ because $1/N \leq \frac{1}{2}$, and $P_k v_0 \neq 0$, so $\hat{u}_N(k) \neq 0$. $\square$

The exact laws rest on a band-symmetry identity which, because χ is not assumed monotone and the truncation is not a sharp ball, must be available for a *general* radial weight, with no positivity assumption.

Lemma 3 (band symmetry, lattice, general radial weight). *Let $f : \mathbb{Z}^3 \setminus \{0\} \rightarrow \mathbb{R}$ be radial, i.e. $f(k) = \varphi(|k|^2)$ for some φ , and suppose $\sum_{k \neq 0} |k|^2 |f(k)| < \infty$. Then for all i, j*

$$\sum_{k \neq 0} k_i k_j f(k) = \delta_{ij} \frac{1}{3} \sum_{k \neq 0} |k|^2 f(k).$$

Proof. Absolute summability of $|k|^2 f$ dominates every summand $|k_i k_j f(k)| \leq |k|^2 |f(k)|$, so all rearrangements below are legitimate. The index set $\mathbb{Z}^3 \setminus \{0\}$ is invariant under each sign flip $\sigma_i : k_i \mapsto -k_i$ and under all coordinate permutations, and f , depending on $|k|^2$ only, is invariant under both. For $i \neq j$ the bijection σ_i negates the summand $k_i k_j f(k)$, so that sum equals its own negative and vanishes. For $i = j$, coordinate permutations carry the three sums $\sum_k k_i^2 f(k)$ into one another, so they are equal; their sum is $\sum_k |k|^2 f(k)$. $\square$

Lemma 4 (band symmetry, continuum). *Let $g : \mathbb{R}^3 \rightarrow \mathbb{R}$ be radial with $\int_{\mathbb{R}^3} |\xi|^2 |g(\xi)| d\xi < \infty$. Then $\int \xi_i \xi_j g d\xi = \delta_{ij} \frac{1}{3} \int |\xi|^2 g d\xi$.*

Proof. Identical: Lebesgue measure on $\mathbb{R}^3$ and g are both invariant under sign flips and permutations of the coordinates, and the hypothesis dominates the integrand. $\square$

Theorem 4 (exact family laws). *Set*

$$S_N^\chi := \sum_{k \neq 0} \frac{\chi(|k|/N)^2}{|k|^2}, \quad T_N^\chi := \sum_{k \neq 0} \frac{\chi(|k|/N)^2}{|k|^4}, \quad \Sigma_N^\chi := \sum_{k \neq 0} \frac{\chi(|k|/N)}{|k|^2},$$

all three finite sums. Then

$$H_0(u_N) = \frac{2}{3} \|v_0\|^2 T_N^\chi, \quad H_1(u_N) = \frac{2}{3} \|v_0\|^2 S_N^\chi, \quad N_0^2(u_N) = \frac{S_N^\chi}{T_N^\chi}, \quad u_N(0) = \frac{2}{3} \Sigma_N^\chi v_0,$$

and consequently $\|u_N\|_\infty^2 / H_1(u_N) \geq 2(\Sigma_N^\chi)^2 / (3S_N^\chi)$.

Proof. Since P_k is an orthogonal projection, $|P_k v_0|^2 = \|v_0\|^2 - (k \cdot v_0)^2 / |k|^2$. Hence

$$H_0 = \sum_{k \neq 0} \frac{\chi(|k|/N)^2}{|k|^4} |P_k v_0|^2 = \|v_0\|^2 \sum_{k \neq 0} \frac{\chi^2}{|k|^4} - \sum_{k \neq 0} \frac{\chi^2 (k \cdot v_0)^2}{|k|^6}.$$

Apply Lemma 3 with the radial weight $f(k) = \chi(|k|/N)^2 |k|^{-6}$ (a finite sum, so the hypothesis is trivially met):

$$\sum_{k \neq 0} \frac{\chi^2 (k \cdot v_0)^2}{|k|^6} = v_{0i} v_{0j} \sum_{k \neq 0} k_i k_j f(k) = \frac{\|v_0\|^2}{3} \sum_{k \neq 0} \frac{\chi^2}{|k|^4} = \frac{\|v_0\|^2}{3} T_N^\chi.$$

Therefore $H_0 = \|v_0\|^2 T_N^\chi - \frac{1}{3} \|v_0\|^2 T_N^\chi = \frac{2}{3} \|v_0\|^2 T_N^\chi$. The same computation with $f = \chi^2 |k|^{-4}$ applied to $H_1 = \sum_k |k|^2 |\hat{u}_N(k)|^2$ gives $H_1 = \frac{2}{3} \|v_0\|^2 S_N^\chi$, and $N_0^2 = H_1 / H_0$ follows. For the point value,

$$u_N(0) = \sum_{k \neq 0} \hat{u}_N(k) = v_0 \Sigma_N^\chi - \sum_{k \neq 0} \frac{\chi k (k \cdot v_0)}{|k|^4},$$

and Lemma 3 with $f = \chi |k|^{-4}$ turns the last sum into $\frac{1}{3} v_0 \Sigma_N^\chi$; hence $u_N(0) = \frac{2}{3} \Sigma_N^\chi v_0$. Finally $\|u_N\|_\infty \geq |u_N(0)| = \frac{2}{3} \Sigma_N^\chi \|v_0\|$, so $\|u_N\|_\infty^2 / H_1 \geq \frac{4}{9} (\Sigma_N^\chi)^2 \|v_0\|^2 / (\frac{2}{3} \|v_0\|^2 S_N^\chi) = \frac{2}{3} (\Sigma_N^\chi)^2 / S_N^\chi$. $\square$

B.3 Two-sided lattice bounds for the bandwidth

The constants below are crude and explicit; no sharpness is attempted.

Lemma 5 (dyadic shell count). *For every integer $j \geq 0$, $\#\{k \in \mathbb{Z}^3 : 2^j \leq |k| < 2^{j+1}\} \leq 64 \cdot 8^j$.*

Proof. Such k satisfy $|k|_\infty \leq |k| < 2^{j+1}$, hence $|k_i| \leq 2^{j+1} - 1$ for each i , so the count is at most $(2^{j+2} - 1)^3 < 2^{3j+6} = 64 \cdot 8^j$. $\square$

Lemma 6 (upper bounds). *With $S_M := \sum_{1 \leq |k| \leq M} |k|^{-2}$ and $T_\infty := \sum_{k \neq 0} |k|^{-4}$ one has $S_M \leq 128M$ for every real $M \geq 1$, and $T_\infty \leq 128$.*

Proof. Let $J = \lfloor \log_2 M \rfloor$. Every k with $1 \leq |k| \leq M$ lies in a shell $2^j \leq |k| < 2^{j+1}$ with $0 \leq j \leq J$, where $|k|^{-2} \leq 4^{-j}$. By Lemma 5, $S_M \leq \sum_{j=0}^J 64 \cdot 8^j 4^{-j} = 64(2^{J+1} - 1) < 128 \cdot 2^J \leq 128M$. Similarly $T_\infty \leq \sum_{j \geq 0} 64 \cdot 8^j 16^{-j} = 64 \cdot 2 = 128$. $\square$

Lemma 7 (lower bound). *For every real $M \geq 16$ one has $S_M \geq M/87$.*

Proof. Consider $\mathcal{B} := \{k \in \mathbb{Z}^3 : \frac{M}{2\sqrt{3}} < k_i \leq \frac{M}{\sqrt{3}} \text{ for } i = 1, 2, 3\}$. For $k \in \mathcal{B}$ we have $|k|^2 \leq 3(M/\sqrt{3})^2 = M^2$ and $|k|^2 > 3M^2/12 = M^2/4$, so every $k \in \mathcal{B}$ is counted in S_M (in particular $k \neq 0$) and contributes $|k|^{-2} \geq M^{-2}$. Each coordinate ranges over a half-open interval of length $M/(2\sqrt{3})$, which contains at least $\frac{M}{2\sqrt{3}} - 1$ integers; for $M \geq 16$, $\frac{M}{2\sqrt{3}} - 1 \geq 0.28868M - 0.0625M \geq 0.226M$. Hence $\#\mathcal{B} \geq (0.226M)^3 > 0.01154M^3$ and $S_M \geq 0.01154M^3 \cdot M^{-2} = 0.01154M \geq M/87$. $\square$

Proposition 1 (two-sided bandwidth bounds). *For every admissible χ , every $v_0 \neq 0$ and every integer $N \geq 32$,*

$$\frac{N}{348} \leq S_N^\chi \leq 128N, \quad 6 \leq T_N^\chi \leq 128, \quad \frac{N}{44544} \leq N_0^2(u_N) \leq \frac{64}{3}N.$$

Proof. Upper bounds: $0 \leq \chi \leq 1$ and $\chi(|k|/N) = 0$ for $|k| > N$, so $S_N^\chi \leq S_N \leq 128N$ and $T_N^\chi \leq T_\infty \leq 128$ by Lemma 6. Lower bound for T_N^χ : for $N \geq 2$ one has $\chi(1/N) = 1$, and the six lattice points with $|k| = 1$ each contribute 1, so $T_N^\chi \geq 6$. Lower bound for S_N^χ : $\chi(|k|/N) = 1$ whenever $|k| \leq N/2$, hence $S_N^\chi \geq S_{N/2}$; for $N \geq 32$ we have $N/2 \geq 16$, so Lemma 7 gives $S_{N/2}^\chi \geq (N/2)/87 \geq N/348$. Combining with $N_0^2 = S_N^\chi/T_N^\chi$ (Theorem 4) yields $N_0^2 \geq \frac{N/348}{128} = \frac{N}{44544}$ and $N_0^2 \leq \frac{128N}{6} = \frac{64}{3}N$. $\square$

B.4 The profile V

Definition 4. On $\mathbb{R}^3$ put

$$h(\xi) := \frac{P_\xi v_0}{|\xi|^2} \quad (\xi \neq 0), \quad F(\xi) := \chi(|\xi|) h(\xi) \quad (\xi \neq 0), \quad F(0) := 0,$$

$$V(y) := \tilde{F}(y) = \int_{\mathbb{R}^3} F(\xi) e^{i\xi \cdot y} d\xi.$$

Because P_ξ is homogeneous of degree 0 we have $P_{k/N} = P_k$ and therefore $F(k/N) = \chi(|k|/N) N^2 P_k v_0 / |k|^2$, i.e. the exact scaling identity

$$\hat{u}_N(k) = N^{-2} F(k/N) \quad \text{for every } k \in \mathbb{Z}^3 \quad (3)$$

(the case $k = 0$ holds by the conventions $\hat{u}_N(0) = 0 = F(0)$). Identity (3) is the engine of the whole argument: it says that the family is a single fixed continuum profile sampled on the lattice $N^{-1}\mathbb{Z}^3$.

Theorem 5 (elementary properties of V). (1) $|F(\xi)| \leq \|v_0\| |\xi|^{-2} \mathbf{1}_{\{|\xi| \leq 1\}}$; hence $F \in L^1(\mathbb{R}^3) \cap L^p(\mathbb{R}^3)$ for every $p < 3/2$, with $\|F\|_{L^1} \leq 4\pi\|v_0\|$.

(2) V is real, even, bounded and continuous, with $\|V\|_\infty \leq 4\pi\|v_0\|$.

(3) $V \in C^\infty(\mathbb{R}^3)$, and for every multi-index β

$$\partial^\beta V(y) = \int_{\mathbb{R}^3} (i\xi)^\beta F(\xi) e^{i\xi \cdot y} d\xi, \quad \|\partial^\beta V\|_\infty \leq \|v_0\| \int_{|\xi| \leq 1} |\xi|^{|\beta|-2} d\xi = \frac{4\pi\|v_0\|}{|\beta| + 1}.$$

In particular each $\|\partial_i V_j\|_\infty \leq 2\pi\|v_0\|$, so $\|\nabla V\|_\infty \leq 6\pi\|v_0\|$ for the Frobenius norm. Moreover V extends to an entire function on $\mathbb{C}^3$ of exponential type ≤ 1 .

(4) $\nabla \cdot V \equiv 0$ on $\mathbb{R}^3$.

(5) $V(0) = \frac{8\pi}{3} \left(\int_0^\infty \chi(r) dr \right) v_0 \neq 0$.

Proof. (1) $|P_\xi v_0| \leq \|v_0\|$, $0 \leq \chi \leq 1$ and $\text{supp } \chi \subset [0, 1]$ give the pointwise bound; $\int_{|\xi| \leq 1} |\xi|^{-2} d\xi = 4\pi \int_0^1 dr = 4\pi$, and $\int_{|\xi| \leq 1} |\xi|^{-2p} d\xi < \infty$ exactly when $2p < 3$.

(2) $F \in L^1$ gives, by dominated convergence, continuity of V and $\|V\|_\infty \leq \|F\|_{L^1} \leq 4\pi\|v_0\|$. F is real-valued and even ($P_{-\xi} = P_\xi$), so $V(y) = \int F(\xi) \cos(\xi \cdot y) d\xi$ is real and even.

(3) By (1), $\xi^\beta F \in L^1$ for every β , since the exponent $|\beta| - 2 > -3$ is locally integrable in $\mathbb{R}^3$ and the support is bounded. Differentiation under the integral sign is then justified by dominated convergence, giving the displayed formula and, by (1) again, the bound $\|v_0\| \int_{|\xi| \leq 1} |\xi|^{|\beta|-2} d\xi = 4\pi\|v_0\| \int_0^1 r^{|\beta|} dr = 4\pi\|v_0\|/(|\beta| + 1)$. Since F is an L^1 function supported in $\{|\xi| \leq 1\}$, the integral $\int F(\xi) e^{i\xi \cdot y} d\xi$ converges for all complex y and is entire in y , with $|V(y)| \leq \|F\|_{L^1} e^{|\text{Im } y|}$; this is exponential type ≤ 1 .

(4) $\nabla \cdot V(y) = \int i \xi \cdot F(\xi) e^{i\xi \cdot y} d\xi$ and $\xi \cdot F(\xi) = \chi(|\xi|) \xi \cdot P_\xi v_0 / |\xi|^2 = 0$ pointwise.

(5) $V(0) = \int F = v_0 \int \frac{\chi(|\xi|)}{|\xi|^2} d\xi - \int \frac{\chi(|\xi|) \xi (\xi \cdot v_0)}{|\xi|^4} d\xi$. Lemma 4 with the radial $g(\xi) = \chi(|\xi|) |\xi|^{-4}$ (integrable against $|\xi|^2$ by (1)) gives $\int \xi_i \xi_j g = \frac{\delta_{ij}}{3} \int |\xi|^2 g = \frac{\delta_{ij}}{3} \int \chi(|\xi|) |\xi|^{-2} d\xi$, so the second integral equals $\frac{1}{3} v_0 \int \chi(|\xi|) |\xi|^{-2} d\xi$. Hence

$$V(0) = \frac{2}{3} v_0 \int_{\mathbb{R}^3} \frac{\chi(|\xi|)}{|\xi|^2} d\xi = \frac{2}{3} v_0 \cdot 4\pi \int_0^\infty \chi(r) dr = \frac{8\pi}{3} \left(\int_0^\infty \chi \right) v_0,$$

and $\int_0^\infty \chi \geq \int_0^{1/2} 1 = \frac{1}{2} > 0$ because $\chi \geq 0$ and $\chi \equiv 1$ on $[0, \frac{1}{2}]$. $\square$

Remark 5 (structural form). $F = P_\xi(v_0 \chi(|\xi|) |\xi|^{-2})$, so $V = \mathbb{P}(v_0 g_\chi)$ where g_χ is the inverse transform of the smoothed Newtonian symbol $\chi(|\cdot|) |\cdot|^{-2}$; solving $\Delta \phi = g_\chi$ with ϕ radial exhibits V in the Oseen normal form $V(y) = a(r) v_0 + b(r) (v_0 \cdot \hat{y}) \hat{y}$ with $a = \phi'' + \phi'/r$, $b = \phi'/r - \phi''$. This form is not used below; it explains why the far field of V is exactly the Oseen profile (Theorem 6).

We record here the two classical transform facts used later.

Remark 6 (classical input: Riesz-potential transforms). *For $0 < \alpha < 3$, and by analytic continuation for every α at which the Gamma factors have no pole, the homogeneous function $|y|^{-\alpha}$ on $\mathbb{R}^3$ is a tempered distribution whose Fourier transform is the homogeneous distribution*

$$\mathcal{F}[|y|^{-\alpha}](\zeta) = c_{3,\alpha}|\zeta|^{\alpha-3}, \quad c_{3,\alpha} = \frac{2^{3-\alpha}\pi^{3/2}\Gamma(\frac{3-\alpha}{2})}{\Gamma(\frac{\alpha}{2})}.$$

See, e.g., [16, 15, 17] (we give no section locators, not having verified them). We use $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, $\Gamma(-\frac{1}{2}) = -2\sqrt{\pi}$, $\Gamma(-\frac{3}{2}) = \frac{4}{3}\sqrt{\pi}$, which give

$$c_{3,-1} = -8\pi, \quad c_{3,1} = 4\pi, \quad c_{3,2} = 2\pi^2, \quad c_{3,4} = -\pi^2, \quad c_{3,6} = \frac{\pi^2}{12},$$

together with the elementary rules $\mathcal{F}[y_a y_b f] = -\partial_a \partial_b \mathcal{F}[f]$ and $\mathcal{F}[y_a y_b y_c y_d f] = \partial_a \partial_b \partial_c \partial_d \mathcal{F}[f]$, valid for tempered f .

Remark 7 (classical input: products and convolutions of homogeneous distributions). *Let $A, B \in L_{\text{loc}}^1(\mathbb{R}^3)$ be homogeneous of degrees $-a, -b$ with $0 < a, b < 3$ and $a + b < 3$, and smooth away from the origin. Then AB is locally integrable and homogeneous of degree $-(a+b)$, hence tempered; $\mathcal{F}A, \mathcal{F}B$ are locally integrable, homogeneous of degrees $a-3, b-3$ and smooth away from the origin; the convolution $\mathcal{F}A * \mathcal{F}B$ converges absolutely at every $\zeta \neq 0$ (because $(3-a) + (3-b) > 3$) and defines a locally integrable homogeneous function of degree $3-a-b$; and the exchange formula*

$$\mathcal{F}[AB] = (2\pi)^{-3} \mathcal{F}A * \mathcal{F}B$$

holds as tempered distributions. See, e.g., [15, 17]. We use this only in Theorem 9, with $a = b = 1$.

B.5 The far field, and why periodisation is unavailable

Theorem 6 (far-field asymptotics). *Let*

$$W(y) := \frac{v_0}{|y|} + \frac{(v_0 \cdot y)y}{|y|^3} \quad (y \neq 0).$$

Then $\tilde{h} = \pi^2 W$ as tempered distributions, and

$$V(y) = \pi^2 W(y) + \rho(y), \quad |\rho(y)| \leq C_M(\chi, v_0)|y|^{-M} \quad \text{for } |y| \geq 1 \text{ and every } 2 \leq M \leq 4.$$

(Four derivatives are all that Definition 2 supplies, and $M = 4$ is all that Remark 8 consumes; if $\chi \in C^m$ with $m \geq 2$ the same proof gives every $2 \leq M \leq m$.)

Proof. (a) *The identity $\tilde{h} = \pi^2 W$. Let $\mathcal{O}_{ij}(y) = \frac{1}{8\pi}(\frac{\delta_{ij}}{|y|} + \frac{y_i y_j}{|y|^3})$ be the Oseen tensor, i.e. the fundamental solution of the steady Stokes system [13, 14], and $h_{ij}(\xi) = \frac{\delta_{ij}}{|\xi|^2} - \frac{\xi_i \xi_j}{|\xi|^4}$, so that $h_i = h_{ij} v_{0j}$. We verify $\mathcal{F}[\mathcal{O}_{ij}] = h_{ij}$ from Remark 6. First, $\mathcal{F}[|y|^{-1}] = c_{3,1}|\zeta|^{-2} = 4\pi|\zeta|^{-2}$, so $\mathcal{F}[\frac{\delta_{ij}}{8\pi|y|}] = \frac{\delta_{ij}}{2|\zeta|^2}$. Second, differentiating $\partial_i |y| = y_i/|y|$ once more gives the pointwise identity*

$$\partial_i \partial_j |y| = \frac{\delta_{ij}}{|y|} - \frac{y_i y_j}{|y|^3}, \quad \text{i.e.} \quad \frac{y_i y_j}{|y|^3} = \frac{\delta_{ij}}{|y|} - \partial_i \partial_j |y|.$$

With $\mathcal{F}[|y|] = c_{3,-1}|\zeta|^{-4} = -8\pi|\zeta|^{-4}$ we get $\mathcal{F}[\partial_i\partial_j|y|] = (i\zeta_i)(i\zeta_j)\mathcal{F}[|y|] = 8\pi\zeta_i\zeta_j|\zeta|^{-4}$, hence

$$\mathcal{F}\left[\frac{y_i y_j}{|y|^3}\right] = \frac{4\pi\delta_{ij}}{|\zeta|^2} - \frac{8\pi\zeta_i\zeta_j}{|\zeta|^4}.$$

Therefore $\mathcal{F}[\mathcal{O}_{ij}] = \frac{1}{8\pi}\left(\frac{4\pi\delta_{ij}}{|\zeta|^2} + \frac{4\pi\delta_{ij}}{|\zeta|^2} - \frac{8\pi\zeta_i\zeta_j}{|\zeta|^4}\right) = \frac{\delta_{ij}}{|\zeta|^2} - \frac{\zeta_i\zeta_j}{|\zeta|^4} = h_{ij}$. All these are identities between homogeneous tempered distributions; possible ambiguities are supported at the origin only and are excluded because both sides are homogeneous of degree $-2 > -3$, hence locally integrable, with no room for a delta term. Fourier inversion ($\mathcal{F}\tilde{G} = (2\pi)^3 G$) now gives $\tilde{h} = (2\pi)^3 \mathcal{O} v_0 = \frac{(2\pi)^3}{8\pi} W = \pi^2 W$, where we used $8\pi \mathcal{O}_{ij} v_{0j} = W_i$.

(b) *The remainder.* Write $G := (1 - \chi(|\cdot|))h$, so that $F - h = -G$ and $\rho := -\tilde{G}$. Then G vanishes on $\{|\xi| < \frac{1}{2}\}$ and is C^4 on all of $\mathbb{R}^3$; since h is homogeneous of degree -2 and smooth off the origin, and $1 - \chi(|\cdot|)$ is C^4 with its first four derivatives bounded, we get

$$|\partial^\beta G(\xi)| \leq C_\beta(\chi)\|v_0\| \langle \xi \rangle^{-2-|\beta|} \quad (\xi \in \mathbb{R}^3, |\beta| \leq 4),$$

which for $2 \leq |\beta| \leq 4$ places $\partial^\beta G$ in $L^1(\mathbb{R}^3)$ (the exponent $2 + |\beta| > 3$). From the distributional identity $y^\beta \tilde{G} = i^{|\beta|} \widetilde{\partial^\beta G}$, the right-hand side for $2 \leq |\beta| \leq 4$ is a bounded continuous function with supremum at most $\|\partial^\beta G\|_{L^1}$. Choosing $\beta = (M, 0, 0), (0, M, 0), (0, 0, M)$ for a given $2 \leq M \leq 4$ and using $|y|^M \leq 3^{M/2} \max_i |y_i|^M$ shows that $\tilde{G}$ agrees on $\{|y| \geq 1\}$ with a continuous function obeying $|\tilde{G}(y)| \leq C_M(\chi, v_0)|y|^{-M}$. Since $V = \tilde{F} = \tilde{h} - \tilde{G} = \pi^2 W + \rho$, the claim follows. $\square$

Thus the decay of V is *exactly* $|y|^{-1}$, with a χ -independent leading profile $\pi^2 W$ satisfying

$$v_0 \cdot W(y) = \frac{\|v_0\|^2}{|y|} + \frac{(v_0 \cdot y)^2}{|y|^3} \geq \frac{\|v_0\|^2}{|y|} > 0. \quad (4)$$

Remark 8 (the naive periodisation argument is unavailable). *It is tempting to reconstruct u_N from V by Poisson summation, $u_N(x) \stackrel{?}{=} N \sum_{m \in \mathbb{Z}^3} V(N(x + 2\pi m))$ with the $m \neq 0$ terms uniformly $O(1/N)$. **The naive absolutely convergent periodisation argument is unavailable:** the series of translates diverges, because V decays only like $|y|^{-1}$ and its leading profile has a sign-definite spherical mean in the v_0 direction, so there is no cancellation to exploit and no absolute convergence to appeal to. Indeed, by Theorem 6 with $M = 4$, for $m \neq 0$ and $x \in [-\pi, \pi)^3$ one has $|N(x + 2\pi m)| \geq N\pi \geq 1$ and therefore, using (4),*

$$v_0 \cdot V(N(x + 2\pi m)) \geq \frac{\pi^2 \|v_0\|^2}{N|x + 2\pi m|} - \frac{C_4 \|v_0\|}{N^4 |x + 2\pi m|^4}.$$

The shell $|m|_\infty = n$ contains $O(n^2)$ points with $|x + 2\pi m| \asymp n$, so $\sum_{m \neq 0} |x + 2\pi m|^{-4} < \infty$ while $\sum_{m \neq 0} |x + 2\pi m|^{-1} = \infty$; hence $\sum_{m \neq 0} v_0 \cdot V(N(x + 2\pi m)) = +\infty$ for every x and every N . The terms are moreover eventually positive, so the divergence is not of a kind that a regular linear summation method would remove: for $a_m \geq 0$ with $\sum a_m = \infty$, monotone convergence gives $\lim_{\varepsilon \downarrow 0} \sum_m e^{-\varepsilon|m|} a_m = +\infty$, so Abel summation in particular does not converge here. We make no assertion beyond this, and none is needed: the point is only that the naive absolutely convergent periodisation argument is not available, because V decays only like $|y|^{-1}$ with a non-oscillating leading profile whose spherical mean, $\frac{4}{3}v_0/r$, is sign-definite in the v_0 direction. If a periodisation formula is wanted purely for orientation, the correct renormalised one,

$$u_N(x) = N \left[V(Nx) + \sum_{m \neq 0} (V(N(x + 2\pi m)) - V(2\pi Nm)) \right] + \text{const},$$

does converge absolutely, the bracketed differences being $O(|m|^{-2})$ by Theorem 6. **Nothing in this appendix uses any periodisation identity:** §B.6 replaces it by an exact lattice-Riemann-sum identity, which is elementary, quantitative, and additionally delivers the derivative statement. This remark, and with it Theorem 6, is inert for the proof of Theorem 2.

B.6 The inner rescaling: an exact lattice-Riemann identity with an explicit rate

For $\Phi : \mathbb{R}^3 \rightarrow \mathbb{C}^d$ put

$$\mathcal{R}_N[\Phi] := N^{-3} \sum_{k \in \mathbb{Z}^3} \Phi(k/N) - \int_{\mathbb{R}^3} \Phi(\xi) d\xi,$$

whenever both terms converge absolutely.

Lemma 8 (exact inner rescaling). *For every $y \in \mathbb{R}^3$ and every integer $N \geq 2$,*

$$u_N(y/N) = N(V(y) + E_N(y)), \quad \partial_i u_{N,j}(y/N) = N^2(\partial_i V_j(y) + E'_{N,ij}(y)),$$

where, with $\Phi_y(\xi) := F(\xi)e^{i\xi \cdot y}$ and $\Phi'_{y,ij}(\xi) := i\xi_i F_j(\xi)e^{i\xi \cdot y}$ (both set to 0 at $\xi = 0$),

$$E_N(y) = \mathcal{R}_N[\Phi_y], \quad E'_{N,ij}(y) = \mathcal{R}_N[\Phi'_{y,ij}].$$

These are identities, not approximations.

Proof. Both series are finite (support $|k| \leq N$) and both integrals converge absolutely by Theorem 5(1),(3). By (3),

$$u_N(y/N) = \sum_{k \in \mathbb{Z}^3} \hat{u}_N(k) e^{i(k/N) \cdot y} = \sum_k N^{-2} F(k/N) e^{i(k/N) \cdot y} = N \cdot N^{-3} \sum_k \Phi_y(k/N),$$

which equals $N(\int \Phi_y + \mathcal{R}_N[\Phi_y]) = N(V(y) + E_N(y))$, using $V(y) = \int \Phi_y$ (Definition 4) and $\Phi_y(0) = F(0) = 0$. For the gradient, $\partial_i u_{N,j}(x) = \sum_k i k_i \hat{u}_{N,j}(k) e^{ik \cdot x}$; at $x = y/N$ this is

$$\sum_k i N^{-2} k_i F_j(k/N) e^{i(k/N) \cdot y} = N^{-1} \sum_k i(k_i/N) F_j(k/N) e^{i(k/N) \cdot y} = N^2 \cdot N^{-3} \sum_k \Phi'_{y,ij}(k/N),$$

and $\int \Phi'_{y,ij} = \partial_i V_j(y)$ by Theorem 5(3). $\square$

Everything now rests on bounding a Riemann-sum defect for an explicit, compactly supported integrand with a single integrable algebraic singularity.

Theorem 7 (C_{loc}^1 convergence with explicit rate). *There is an absolute constant A (the proof gives $A = 3000$) such that for every real $R \geq 1$ and every integer $N \geq 8R$,*

$$\varepsilon_N := \sup_{|y| \leq R} |E_N(y)| \leq A \|v_0\| \frac{c_\chi(1 + \log_2 N) + R}{N}, \quad \varepsilon'_N := \sup_{|y| \leq R} |E'_{N,ij}(y)| \leq A \|v_0\| \frac{c_\chi + R}{N}.$$

In particular $\varepsilon_N + \varepsilon'_N = O(N^{-1} \log N) \rightarrow 0$ as $N \rightarrow \infty$, for each fixed R .

Proof. Partition $\mathbb{R}^3$ into the half-open cubes $Q_k = \frac{k}{N} + [0, \frac{1}{N})^3$, $k \in \mathbb{Z}^3$, of side $1/N$, volume N^{-3} and diameter $\sqrt{3}/N$. For any Φ for which both the sum and the integral converge absolutely,

$$\mathcal{R}_N[\Phi] = \sum_{k \in \mathbb{Z}^3} \int_{Q_k} [\Phi(k/N) - \Phi(\xi)] d\xi, \quad (5)$$

since $|Q_k| = N^{-3}$. Absolute convergence holds for $\Phi = \Phi_y$ and $\Phi = \Phi'_{y,ij}$: both are supported in $\{|\xi| \leq 1\}$ and dominated there by $\|v_0\||\xi|^{-2}$ respectively $\|v_0\||\xi|^{-1}$, which are integrable on $\mathbb{R}^3$; and the lattice sums are finite sums, bounded by $N^{-3} \sum_{0 < |k| \leq N} \|v_0\| N^2 |k|^{-2} = \|v_0\| N^{-1} S_N \leq 128\|v_0\|$ and by $N^{-3} \sum_{0 < |k| \leq N} \|v_0\| N |k|^{-1} \leq \|v_0\| N^{-2} \cdot N S_N \leq 128\|v_0\|$ respectively, using $|k|^{-1} \leq N|k|^{-2}$ for $|k| \leq N$ and Lemma 6.

Pointwise bounds. On $0 < |\xi| \leq 1$ we have $|F(\xi)| \leq \|v_0\||\xi|^{-2}$, and by the product rule applied to $F = \chi(|\xi|)|\xi|^{-2} P_\xi v_0$,

$$|\nabla F(\xi)| \leq \underbrace{c_\chi \|v_0\||\xi|^{-2}}_{\nabla \chi(|\xi|)} + \underbrace{2\|v_0\||\xi|^{-3}}_{\chi \nabla(|\xi|^{-2})} + \underbrace{4\|v_0\||\xi|^{-3}}_{\chi |\xi|^{-2} \nabla(P_\xi v_0)} \leq 7c_\chi \|v_0\| |\xi|^{-3},$$

where we used $c_\chi \geq 1$, $|\xi| \leq 1$, $|\nabla(|\xi|^{-2})| = 2|\xi|^{-3}$, and, from $\partial_a(\xi_i \xi_j / |\xi|^2) = (\delta_{ai} \xi_j + \delta_{aj} \xi_i) / |\xi|^2 - 2\xi_i \xi_j \xi_a / |\xi|^4$, the honest bound $|\nabla(P_\xi v_0)| \leq 4\|v_0\|/|\xi|$. Consequently, for $|y| \leq R$,

$$|\Phi_y| \leq \frac{\|v_0\|}{|\xi|^2}, \quad |\nabla \Phi_y| \leq \frac{7c_\chi \|v_0\|}{|\xi|^3} + \frac{R\|v_0\|}{|\xi|^2}, \quad |\Phi'_y| \leq \frac{\|v_0\|}{|\xi|}, \quad |\nabla \Phi'_y| \leq \frac{8c_\chi \|v_0\|}{|\xi|^2} + \frac{R\|v_0\|}{|\xi|}. \quad (6)$$

(For the last one, $\nabla(i\xi_i F_j e^{i\xi y})$ has the three terms $|F| \leq \|v_0\||\xi|^{-2}$, $|\xi||\nabla F| \leq 7c_\chi \|v_0\||\xi|^{-2}$ and $R|\xi||F| \leq R\|v_0\||\xi|^{-1}$.)

The core. Let $\mathcal{K}_0 := \{k : Q_k \cap \{|\xi| \leq 4\sqrt{3}/N\} \neq \emptyset\}$. Any such Q_k lies in $\{|\xi| \leq 5\sqrt{3}/N\}$ and has $|k| \leq 5\sqrt{3} < 9$, i.e. $|k| \leq 8$. Note that $k = 0$ is in the core, and that although $\Phi_y(0) = 0$ by convention while $\int_{Q_0} \Phi_y \neq 0$, both are bounded by the two terms below, so no separate treatment is needed. The core contributes to (5) at most

$$\int_{|\xi| \leq 5\sqrt{3}/N} |\Phi_y| d\xi + N^{-3} \sum_{0 < |k| \leq 8} |\Phi_y(k/N)| \leq \|v_0\| \left(\frac{4\pi \cdot 5\sqrt{3}}{N} + \frac{1}{N} \sum_{0 < |k| \leq 8} \frac{1}{|k|^2} \right) \leq \frac{1133\|v_0\|}{N},$$

using $\int_{|\xi| \leq a} |\xi|^{-2} d\xi = 4\pi a$, then $|\Phi_y(k/N)| \leq \|v_0\| N^2 |k|^{-2}$, and finally $\sum_{0 < |k| \leq 8} |k|^{-2} \leq S_8 \leq 1024$ by Lemma 6 ($4\pi \cdot 5\sqrt{3} \leq 109$, and $109 + 1024 = 1133$).

Dyadic annuli. Put $J := \lfloor \log_2(N/(4\sqrt{3})) \rfloor$, which is ≥ 0 because $N \geq 8R \geq 8 > 4\sqrt{3}$. Let $k \notin \mathcal{K}_0$ and set $r_k := \inf_{\xi \in \overline{Q_k}} |\xi|$, so $r_k > 4\sqrt{3}/N > 0$. If $r_k > 1$ then Φ_y vanishes identically on $\overline{Q_k}$ and that cell contributes 0; otherwise choose the unique integer $j(k)$ with $2^{-j(k)-1} < r_k \leq 2^{-j(k)}$. Then $j(k) \geq 0$, and $2^{-j(k)} \geq r_k > 4\sqrt{3}/N$ forces $2^{j(k)} < N/(4\sqrt{3})$, i.e. $0 \leq j(k) \leq J$; so the annuli $A_j := \{2^{-j-1} < |\xi| \leq 2^{-j}\}$, $0 \leq j \leq J$, together with the core, do cover $\text{supp } \Phi_y$.

Fix j and consider the cells with $j(k) = j$. They are pairwise disjoint. Each satisfies $\overline{Q_k} \subset \{|\xi| \geq 2^{-j-1}\}$ and $\overline{Q_k} \subset \{|\xi| \leq r_k + \sqrt{3}/N \leq \frac{5}{4} 2^{-j}\}$, because $\sqrt{3}/N \leq \frac{1}{4} 2^{-j}$ (which follows from $2^{-j} \geq 2^{-J} \geq 4\sqrt{3}/N$). Hence all of them lie in the fixed annulus $B_j := \{2^{-j-1} \leq |\xi| \leq \frac{5}{4} 2^{-j}\}$, whose volume is $\frac{4\pi}{3} ((\frac{5}{4})^3 - (\frac{1}{2})^3) 2^{-3j} \leq 9 \cdot 2^{-3j}$; therefore $\sum_{j(k)=j} |Q_k| \leq 9 \cdot 2^{-3j}$.

Since $\overline{Q_k}$ is convex, avoids the origin, and Φ_y is C^1 there, the mean value inequality gives $|\int_{Q_k} [\Phi_y(k/N) - \Phi_y(\xi)] d\xi| \leq \frac{\sqrt{3}}{N} \sup_{\overline{Q_k}} |\nabla \Phi_y| \cdot |Q_k|$. By (6) and $|\xi| \geq 2^{-j-1}$ on $\overline{Q_k}$,

$$\sup_{\overline{Q_k}} |\nabla \Phi_y| \leq 7c_\chi \|v_0\| 2^{3(j+1)} + R\|v_0\| 2^{2(j+1)} = 56c_\chi \|v_0\| 8^j + 4R\|v_0\| 4^j.$$

Summing over the cells with $j(k) = j$,

$$\sum_{j(k)=j} \left| \int_{Q_k} [\Phi_y(k/N) - \Phi_y] \right| \leq \frac{\sqrt{3}}{N} (56c_\chi \|v_0\| 8^j + 4R \|v_0\| 4^j) \cdot 9 \cdot 2^{-3j} \leq \frac{\|v_0\|}{N} (873c_\chi + 63R 2^{-j}).$$

Summing over $0 \leq j \leq J \leq \log_2 N$ and adding the core bound,

$$|E_N(y)| \leq \frac{\|v_0\|}{N} (1133 + 873c_\chi(1 + \log_2 N) + 126R) \leq \frac{3000\|v_0\|}{N} (c_\chi(1 + \log_2 N) + R),$$

using $c_\chi \geq 1$ and $1 + \log_2 N \geq 1$. This is the first display.

The gradient. Repeat verbatim with $\Phi'_{y,ij}$. The core contributes at most

$$\int_{|\xi| \leq 5\sqrt{3}/N} \frac{\|v_0\|}{|\xi|} d\xi + N^{-3} \sum_{0 < |k| \leq 8} \frac{\|v_0\| N}{|k|} = \|v_0\| \left(\frac{2\pi \cdot 75}{N^2} + \frac{1}{N^2} \sum_{0 < |k| \leq 8} \frac{1}{|k|} \right) \leq \frac{1816\|v_0\|}{N^2} \leq \frac{1816\|v_0\|}{N},$$

where $\int_{|\xi| \leq a} |\xi|^{-1} d\xi = 2\pi a^2$ and, by Lemma 5, $\sum_{0 < |k| \leq 8} |k|^{-1} \leq \sum_{j=0}^2 64 \cdot 8^j 2^{-j} = 1344$ (and $2\pi \cdot 75 \leq 472$, $472 + 1344 = 1816$). On the annuli, $\sup_{Q_k} |\nabla \Phi'_y| \leq 8c_\chi \|v_0\| 2^{2(j+1)} + R \|v_0\| 2^{j+1} = 32c_\chi \|v_0\| 4^j + 2R \|v_0\| 2^j$, so the level- j contribution is at most

$$\frac{\sqrt{3}}{N} (32c_\chi \|v_0\| 4^j + 2R \|v_0\| 2^j) \cdot 9 \cdot 2^{-3j} \leq \frac{\|v_0\|}{N} (499c_\chi 2^{-j} + 32R 4^{-j}),$$

and now the geometric series converge *without* a logarithm, summing over all $j \geq 0$ to at most $\frac{\|v_0\|}{N} (998c_\chi + 43R)$. Adding the core, $|E'_N(y)| \leq \frac{\|v_0\|}{N} (1816 + 998c_\chi + 43R) \leq \frac{3000\|v_0\|}{N} (c_\chi + R)$. The absence of a logarithm is structural: the singularity of Φ'_y is only $|\xi|^{-1}$, the extra factor ξ taming the origin. The derivative estimate is thus not a consequence of the C^0 one, but it is the easier of the two. $\square$

B.7 The concentrated test field and the duality lower bound

Definition 5. Let $\Psi \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$ be *real*, with $\nabla \cdot \Psi = 0$, $\text{supp } \Psi \subset \{|y| \leq R\}$ for some $R \geq 1$, and $\Psi \not\equiv 0$. For $N > R/\pi$ define on $\mathbb{T}^3$

$$\psi_N(x) := N^{3/2} \sum_{m \in \mathbb{Z}^3} \Psi(N(x + 2\pi m)).$$

Lemma 9 (properties of ψ_N). *Let $N > R/\pi$. Then ψ_N is a well-defined real C^∞ vector field on $\mathbb{T}^3$; it is divergence-free and has zero mean; on the fundamental domain $[-\pi, \pi]^3$ exactly the term $m = 0$ is nonzero, so $\text{supp } \psi_N \subset \{|x| \leq R/N\}$ there; and*

$$\|\psi_N\|_{L^2(\mathbb{T}^3)} = (2\pi)^{-3/2} \|\Psi\|_{L^2(\mathbb{R}^3)},$$

which is independent of N .

Proof. Local finiteness: $\Psi(N(x + 2\pi m)) \neq 0$ forces $|x + 2\pi m| \leq R/N < \pi$; but for $x \in [-\pi, \pi]^3$ and $m \neq 0$, $|x + 2\pi m| \geq 2\pi|m|_\infty - \pi \geq \pi$, a contradiction. Hence on a neighbourhood of any point the sum has at most one nonzero term, so ψ_N is smooth, real, and $2\pi\mathbb{Z}^3$ -periodic by

construction. Divergence-free: locally $\nabla \cdot \psi_N(x) = N^{5/2}(\nabla \cdot \Psi)(N(x + 2\pi m)) = 0$. Zero mean: with $d\mu = (2\pi)^{-3}dx$ and the substitution $y = Nx$,

$$\int_{\mathbb{T}^3} \psi_N d\mu = (2\pi)^{-3} N^{3/2} \int_{\mathbb{R}^3} \Psi(Nx) dx = (2\pi)^{-3} N^{-3/2} \int_{\mathbb{R}^3} \Psi,$$

and $\int_{\mathbb{R}^3} \Psi_i = \int_{\mathbb{R}^3} \nabla \cdot (y_i \Psi) = 0$ since $\nabla \cdot \Psi = 0$ and $y_i \Psi \in C_c^\infty$. (No curl representation of Ψ is needed: zero mean is automatic for a compactly supported divergence-free field.) Norm:

$$\|\psi_N\|_{L^2(\mathbb{T}^3)}^2 = (2\pi)^{-3} \int_{|x| \leq R/N} N^3 |\Psi(Nx)|^2 dx = (2\pi)^{-3} N^3 N^{-3} \|\Psi\|_{L^2(\mathbb{R}^3)}^2,$$

which is the stated identity. $\square$

Theorem 8 (duality identity with controlled error). *Let Ψ be as in Definition 5 and put*

$$I_\Psi := \int_{\mathbb{R}^3} (V \cdot \nabla V) \cdot \Psi dy.$$

Then for every integer $N \geq \max(8R, 32)$,

$$\langle \mathbb{P}(u_N \cdot \nabla u_N), \psi_N \rangle_{L^2(\mathbb{T}^3)} = (2\pi)^{-3} N^{3/2} (I_\Psi + \mathcal{E}_N),$$

where

$$|\mathcal{E}_N| \leq \|\Psi\|_{L^1(\mathbb{R}^3)} \left[\varepsilon_N (6\pi \|v_0\| + \varepsilon'_N) + 4\pi \|v_0\| \varepsilon'_N \right] = O\left(\frac{\log N}{N}\right)$$

with $\varepsilon_N, \varepsilon'_N$ as in Theorem 7.

Proof. By Lemma 9, ψ_N is zero-mean and divergence-free, so $\hat{\psi}_N(0) = 0$ and $k \cdot \hat{\psi}_N(k) = 0$ for all k , whence $\mathbb{P}\psi_N = \psi_N$. Since $\mathbb{P}$ is an orthogonal projection on $L^2(\mathbb{T}^3)$, hence self-adjoint,

$$\langle \mathbb{P}(u_N \cdot \nabla u_N), \psi_N \rangle = \langle u_N \cdot \nabla u_N, \mathbb{P}\psi_N \rangle = \langle u_N \cdot \nabla u_N, \psi_N \rangle.$$

Both fields are real. Because $N \geq 8R$ and $R \geq 1$ we have $R/N \leq \frac{1}{8} < \pi$, so Lemma 9 applies and $\psi_N(x) = N^{3/2}\Psi(Nx)$ on the fundamental domain, supported in $\{|x| \leq R/N\}$. Using $d\mu = (2\pi)^{-3}dx$ and substituting $x = y/N$ ($dx = N^{-3}dy$),

$$\langle u_N \cdot \nabla u_N, \psi_N \rangle = (2\pi)^{-3} N^{3/2} N^{-3} \int_{|y| \leq R} (u_N \cdot \nabla u_N)(y/N) \cdot \Psi(y) dy.$$

By Lemma 8, for $|y| \leq R$,

$$(u_N \cdot \nabla u_N)_j(y/N) = u_{N,i}(y/N) \partial_i u_{N,j}(y/N) = N^3 (V_i(y) + E_{N,i}(y)) (\partial_i V_j(y) + E'_{N,ij}(y)),$$

so the factor $N^{-3}N^3$ cancels and

$$\langle u_N \cdot \nabla u_N, \psi_N \rangle = (2\pi)^{-3} N^{3/2} (I_\Psi + \mathcal{E}_N), \quad \mathcal{E}_N = \int_{|y| \leq R} [E_{N,i} \partial_i V_j + V_i E'_{N,ij} + E_{N,i} E'_{N,ij}] \Psi_j dy,$$

where we used $\text{supp } \Psi \subset \{|y| \leq R\}$ to write the main term as I_Ψ . Bounding each factor by its supremum and using Theorem 5(2),(3) — $\|V\|_\infty \leq 4\pi \|v_0\|$, $\|\nabla V\|_\infty \leq 6\pi \|v_0\|$ — gives the stated estimate; $\varepsilon_N, \varepsilon'_N = O(N^{-1} \log N)$ by Theorem 7, R and Ψ being fixed. $\square$

Proposition 2 (the N^3 lower bound, given a good test field). *Suppose some Ψ as in Definition 5 has $I_\Psi \neq 0$. Then*

$$\|\mathbb{P}(u_N \cdot \nabla u_N)\|_{L^2(\mathbb{T}^3)}^2 \geq \frac{|I_\Psi + \mathcal{E}_N|^2}{(2\pi)^3 \|\Psi\|_{L^2(\mathbb{R}^3)}^2} N^3 \quad (N \geq \max(8R, 32)),$$

and consequently $\|\mathbb{P}(u_N \cdot \nabla u_N)\|_2^2 \geq c_0 N^3$ for all $N \geq N_*$, where

$$c_0 = \frac{|I_\Psi|^2}{2(2\pi)^3 \|\Psi\|_{L^2(\mathbb{R}^3)}^2}, \quad N_* = \min \left\{ N_0 \geq \max(8R, 32) : |\mathcal{E}_N| \leq (1 - 2^{-1/2})|I_\Psi| \quad \forall N \geq N_0 \right\},$$

which is finite because $\mathcal{E}_N \rightarrow 0$.

Proof. By Cauchy–Schwarz in $L^2(\mathbb{T}^3)$,

$$\|\mathbb{P}(u_N \cdot \nabla u_N)\|_2 \geq \frac{|\langle \mathbb{P}(u_N \cdot \nabla u_N), \psi_N \rangle|}{\|\psi_N\|_2}.$$

Insert Theorem 8 and Lemma 9: the right-hand side equals

$$\frac{(2\pi)^{-3} N^{3/2} |I_\Psi + \mathcal{E}_N|}{(2\pi)^{-3/2} \|\Psi\|_{L^2(\mathbb{R}^3)}} = (2\pi)^{-3/2} N^{3/2} \frac{|I_\Psi + \mathcal{E}_N|}{\|\Psi\|_{L^2(\mathbb{R}^3)}}.$$

Squaring gives the display. For $N \geq N_*$ we have $|I_\Psi + \mathcal{E}_N| \geq 2^{-1/2}|I_\Psi|$, whence $|I_\Psi + \mathcal{E}_N|^2 \geq \frac{1}{2}|I_\Psi|^2$ and the constant c_0 as stated. $\square$

Everything is now reduced to the single question of whether some admissible Ψ has $I_\Psi \neq 0$.

B.8 Non-degeneracy of the continuum nonlinearity

Reduction to a symbol statement

Lemma 10 (symbol of the nonlinearity). *$V \cdot \nabla V = \nabla \cdot (V \otimes V)$ has the representation*

$$(V \cdot \nabla V)(y) = \int_{\mathbb{R}^3} M(\zeta) e^{i\zeta \cdot y} d\zeta, \quad M_i(\zeta) = i \zeta_j T_{ij}(\zeta), \quad T_{ij} := F_i * F_j,$$

where T is real, symmetric, supported in $\{|\zeta| \leq 2\}$, and $T, M \in L^1(\mathbb{R}^3) \cap L^2(\mathbb{R}^3)$. Consequently $\mathbb{P}(V \cdot \nabla V) \in L^2(\mathbb{R}^3)$ with symbol $P_\zeta M(\zeta)$, and

$$\mathbb{P}(V \cdot \nabla V) \equiv 0 \iff \zeta \times (T(\zeta)\zeta) = 0 \text{ for a.e. } \zeta.$$

Proof. $V = \widetilde{F}$ with $F \in L^1$, and $\partial_j V_i = \widetilde{i\xi_j F_i}$ with $\xi F \in L^1$ (Theorem 5(1),(3)). The pointwise product of inverse transforms of L^1 functions is the inverse transform of the convolution, so $V_j \partial_j V_i = F_j * \widetilde{(i \cdot_j F_i)}$, i.e. the symbol of $V \cdot \nabla V$ is $i \int F_j(\zeta - \eta) \eta_j F_i(\eta) d\eta$. Since $(\zeta - \eta) \cdot F(\zeta - \eta) = 0$ identically (Definition 4), we may replace $\eta_j F_j(\zeta - \eta)$ by $\zeta_j F_j(\zeta - \eta)$ inside the integral, obtaining $M_i(\zeta) = i \zeta_j (F_i * F_j)(\zeta) = i \zeta_j T_{ij}(\zeta)$.

T is real because F is real, symmetric because convolution is commutative, and supported in $\{|\zeta| \leq 2\}$ because $\text{supp } F \subset \{|\xi| \leq 1\}$. Integrability: by Theorem 5(1), $F \in L^{4/3}$; Young's inequality with $\frac{1}{r} = \frac{2}{4/3} - 1 = \frac{1}{2}$ gives $T = F * F \in L^2$, and since T has compact support, $T \in L^1 \cap L^2$; the same holds for $M = i\zeta T$, as $|\zeta| \leq 2$ on the support. By Plancherel, $V \cdot \nabla V \in L^2(\mathbb{R}^3)$ and, since $|P_\zeta M| \leq |M|$, $\mathbb{P}(V \cdot \nabla V) \in L^2(\mathbb{R}^3)$ with symbol $P_\zeta M$. Finally $\mathbb{P}(V \cdot \nabla V) = 0$ in L^2 iff $P_\zeta M(\zeta) = 0$ for a.e. ζ , and for a vector w one has $P_\zeta w = 0$ iff $w \parallel \zeta$ iff $\zeta \times w = 0$; here $w = i T(\zeta)\zeta$. $\square$

Remark 9 (classical input: density of smooth compactly supported solenoidal fields). *The space $C_{c,\sigma}^\infty(\mathbb{R}^3; \mathbb{R}^3)$ of real, smooth, compactly supported, divergence-free vector fields is dense in $L_\sigma^2(\mathbb{R}^3; \mathbb{R}^3)$, the closed subspace of real L^2 fields that are distributionally divergence-free; indeed L_σ^2 is by the standard definition the L^2 -closure of $C_{c,\sigma}^\infty$, and on $\mathbb{R}^3$ the two descriptions agree. See, e.g., [5, 13, 4].*

Remark 10 (classical input: Helmholtz–Leray decomposition on $\mathbb{R}^3$). *For $f \in L^2(\mathbb{R}^3; \mathbb{R}^3)$ one has $f - \mathbb{P}f = \nabla p$ for some $p \in L_{\text{loc}}^2(\mathbb{R}^3)$ with $\nabla p \in L^2$; equivalently, the symbol of $f - \mathbb{P}f$ is $\zeta(\zeta \cdot \hat{f})/|\zeta|^2$. Same references.*

Lemma 11 (from non-degeneracy to a test field). *If $\mathbb{P}(V \cdot \nabla V) \neq 0$, then there exists a real $\Psi \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$ with $\nabla \cdot \Psi = 0$ and $I_\Psi = \int_{\mathbb{R}^3} (V \cdot \nabla V) \cdot \Psi \neq 0$; after enlarging R if necessary, Ψ satisfies Definition 5.*

Proof. By Lemma 10, $\mathbb{P}(V \cdot \nabla V)$ lies in $L^2(\mathbb{R}^3)$, is real (its symbol $P_\zeta M$ satisfies the reality condition because F is real and even), is distributionally divergence-free (its symbol is orthogonal to ζ), and is nonzero by hypothesis. Hence it is a nonzero element of the real Hilbert space $L_\sigma^2(\mathbb{R}^3)$. By Remark 9 there is a real $\Psi \in C_{c,\sigma}^\infty(\mathbb{R}^3)$ with $\langle \mathbb{P}(V \cdot \nabla V), \Psi \rangle_{L^2(\mathbb{R}^3)} \neq 0$. By Remark 10, $V \cdot \nabla V - \mathbb{P}(V \cdot \nabla V) = \nabla p$ with $\nabla p \in L^2$, and $\int \nabla p \cdot \Psi = -\int p \nabla \cdot \Psi = 0$ because Ψ is compactly supported and divergence-free. Therefore $I_\Psi = \langle \mathbb{P}(V \cdot \nabla V), \Psi \rangle \neq 0$. Choosing $R \geq \max(1, \sup\{|y| : y \in \text{supp } \Psi\})$ puts Ψ in Definition 5. $\square$

The χ -independent leading singularity

By rotational equivariance we may and do assume from now on that $v_0 = \|v_0\|e_3$: F and V are linear in v_0 and rotate with it, T and τ are quadratic, and the conclusion $\mathbb{P}(V \cdot \nabla V) \neq 0$ is rotation invariant.

Definition 6. $\tau_{ij} := h_i * h_j$, with $h(\xi) = P_\xi v_0/|\xi|^2$ the uncut symbol of Definition 4.

Lemma 12 (τ is well defined and homogeneous). *The integral $\tau_{ij}(\zeta) = \int_{\mathbb{R}^3} h_i(\eta)h_j(\zeta - \eta) d\eta$ converges absolutely for every $\zeta \neq 0$, and τ is homogeneous of degree -1 and smooth on $\mathbb{R}^3 \setminus \{0\}$.*

Proof. $|h(\xi)| \leq \|v_0\||\xi|^{-2}$. Near $\eta = 0$ the integrand is $O(|\eta|^{-2})$ and near $\eta = \zeta$ it is $O(|\zeta - \eta|^{-2})$, both locally integrable on $\mathbb{R}^3$; for $|\eta| \geq 2|\zeta|$ it is $O(|\eta|^{-4})$, integrable at infinity. Substituting $\eta = \lambda\eta'$ and using $h(\lambda\eta) = \lambda^{-2}h(\eta)$ gives $\tau(\lambda\zeta) = \lambda^{-1}\tau(\zeta)$ for $\lambda > 0$. Smoothness away from 0 follows from differentiating under the integral sign on the region where the two singularities are separated. $\square$

Theorem 9 (closed form of the leading obstruction). *With $v_0 = \|v_0\|e_3$, for every $\zeta \neq 0$,*

$$\zeta \times (\tau(\zeta)\zeta) = \frac{3\pi^3}{8} \|v_0\|^2 \frac{\zeta_3}{|\zeta|} (\zeta \times e_3).$$

In particular $\zeta \times (\tau(\zeta)\zeta) \neq 0$ whenever $\zeta_3 \neq 0$ and $\zeta \nparallel e_3$.

Proof. Both sides are quadratic in $\|v_0\|$, so we normalise $\|v_0\| = 1$, $v_0 = e_3$. By Theorem 6, $\tilde{h} = \pi^2 W$ with

$$W_i(y) = \frac{\delta_{i3}}{r} + \frac{y_3 y_i}{r^3}, \quad r = |y|.$$

W_i is locally integrable, homogeneous of degree -1 and smooth away from the origin, so Remark 7 applies with $a = b = 1$ and gives

$$\tau_{ij} = h_i * h_j = (2\pi)^{-3} \mathcal{F}[\widetilde{h_i h_j}] = (2\pi)^{-3} \pi^4 \mathcal{F}[W_i W_j],$$

where we used $\mathcal{F}\widetilde{h_i} = (2\pi)^3 h_i$ and the evenness of h . (Both sides are homogeneous of degree -1 and locally integrable, so no distribution supported at the origin can be added.) Expanding,

$$W_i W_j = \frac{\delta_{i3} \delta_{j3}}{r^2} + \frac{\delta_{i3} y_3 y_j + \delta_{j3} y_3 y_i}{r^4} + \frac{y_3^2 y_i y_j}{r^6}.$$

Each bracket is locally integrable and homogeneous of degree -2 , so by Remark 6 its transform is an honest homogeneous distribution of degree -1 , given by

$$\mathcal{F}\left[\frac{1}{r^2}\right] = \frac{2\pi^2}{s}, \quad \mathcal{F}\left[\frac{y_i y_j}{r^4}\right] = \pi^2 \left(\frac{\delta_{ij}}{s} - \frac{\zeta_i \zeta_j}{s^3} \right), \quad \mathcal{F}\left[\frac{y_3^2 y_i y_j}{r^6}\right] = \frac{\pi^2}{12} \partial_3^2 \partial_i \partial_j s^3,$$

where $s := |\zeta|$. (Indeed $\mathcal{F}[r^{-2}] = c_{3,2}s^{-1}$; $\mathcal{F}[y_i y_j r^{-4}] = -\partial_i \partial_j (c_{3,4}s) = \pi^2 \partial_i \partial_j s = \pi^2 (\delta_{ij}/s - \zeta_i \zeta_j/s^3)$; $\mathcal{F}[y_3^2 y_i y_j r^{-6}] = \partial_3^2 \partial_i \partial_j (c_{3,6}s^3)$. Trace check on the middle formula: $\pi^2(3/s - 1/s) = 2\pi^2/s = \mathcal{F}[r^{-2}]$, as it must be.) Direct differentiation of s^3 gives

$$\partial_3^2 \partial_i \partial_j s^3 = 3 \left[\frac{\delta_{ij} + 2\delta_{i3} \delta_{j3}}{s} - \frac{\zeta_3^2 \delta_{ij} + 2\zeta_3 (\delta_{i3} \zeta_j + \delta_{j3} \zeta_i) + \zeta_i \zeta_j}{s^3} + \frac{3\zeta_i \zeta_j \zeta_3^2}{s^5} \right].$$

Assembling the three pieces (the middle one contributing $\pi^2[2\delta_{i3} \delta_{j3}/s - \zeta_3 (\delta_{i3} \zeta_j + \delta_{j3} \zeta_i)/s^3]$) yields $\mathcal{F}[W_i W_j] = \pi^2 G_{ij}$ with

$$G_{ij} = \frac{\delta_{ij}}{4s} + \frac{9\delta_{i3} \delta_{j3}}{2s} - \frac{\zeta_3^2 \delta_{ij}}{4s^3} - \frac{3\zeta_3 (\delta_{i3} \zeta_j + \delta_{j3} \zeta_i)}{2s^3} - \frac{\zeta_i \zeta_j}{4s^3} + \frac{3\zeta_i \zeta_j \zeta_3^2}{4s^5},$$

so that $\tau_{ij} = (2\pi)^{-3} \pi^4 \cdot \pi^2 G_{ij} = \frac{\pi^3}{8} G_{ij}$.

Contract with ζ_j , using $\zeta_j \zeta_j = s^2$. The terms $\frac{\delta_{ij}}{4s} \zeta_j = \frac{\zeta_i}{4s}$ and $-\frac{\zeta_i \zeta_j}{4s^3} \zeta_j = -\frac{\zeta_i}{4s}$ cancel. The $\delta_{i3} \zeta_3/s$ coefficients are $\frac{9}{2}$ from the second term and $-\frac{3}{2}$ from the fourth, total 3. The $\zeta_i \zeta_3^2/s^3$ coefficients are $-\frac{1}{4}$ (third term), $-\frac{3}{2}$ (fourth term) and $+\frac{3}{4}$ (sixth term), total -1 . Hence

$$G(\zeta) \zeta = \frac{3\zeta_3}{s} e_3 - \frac{\zeta_3^2}{s^3} \zeta, \quad \text{so} \quad \zeta \times (G(\zeta) \zeta) = \frac{3\zeta_3}{s} (\zeta \times e_3),$$

the second term being parallel to ζ . Multiplying by $\frac{\pi^3}{8}$ and restoring $\|v_0\|^2$ gives the claim. Finally $\zeta \times e_3 = 0$ exactly when $\zeta \parallel e_3$. $\square$

Remark 11 (physical-space cross-check, free of any regularisation). *Theorem 9 passes through the finite-part continuations of Remark 6. The following elementary computation confirms it independently, with no regularisation and no Fourier transform, and is worth recording because it identifies what the obstruction is: the statement that the Oseen field is not a stationary Euler flow.*

For $W(y) = \frac{e_3}{r} + \frac{y_3 y}{r^3}$ on $\mathbb{R}^3 \setminus \{0\}$,

$$\nabla \cdot W = 0, \quad \omega := \nabla \times W = \frac{2(e_3 \times y)}{r^3}, \quad \nabla \times (W \times \omega) = \frac{12 y_3 (e_3 \times y)}{r^6} \neq 0.$$

Proof. $\nabla \cdot (e_3/r) = e_3 \cdot \nabla r^{-1} = -y_3/r^3$, and $\partial_i(y_3 y_i r^{-3}) = y_3 r^{-3} + 3y_3 r^{-3} - 3y_3 r^{-3} = y_3/r^3$, so $\nabla \cdot W = 0$. Next $\nabla \times (r^{-1}e_3) = \nabla(r^{-1}) \times e_3 = -r^{-3}y \times e_3 = r^{-3}(e_3 \times y)$, while $[\nabla \times (y_3 y r^{-3})]_a = \epsilon_{abi} \partial_b(y_3 y_i r^{-3}) = \epsilon_{abi}(\delta_{b3} y_i + y_3 \delta_{bi}) r^{-3} - 3\epsilon_{abi} y_3 y_i y_b r^{-5} = \epsilon_{a3i} y_i r^{-3} = (e_3 \times y)_a r^{-3}$, the last two terms vanishing by antisymmetry. Adding gives ω . Then, using $e_3 \times (e_3 \times y) = y_3 e_3 - y$ and $y \times (e_3 \times y) = r^2 e_3 - y_3 y$,

$$W \times \omega = \frac{2(y_3 e_3 - y)}{r^4} + \frac{2y_3 e_3}{r^4} - \frac{2y_3^2 y}{r^6} = \frac{4y_3 e_3}{r^4} - \frac{2y}{r^4} - \frac{2y_3^2 y}{r^6}.$$

Now curl each piece. For any radial f , $\nabla \times (f(r)y) = \nabla f \times y = f'(r)\hat{y} \times y = 0$, so $-2y/r^4$ contributes nothing. With $g = -2y_3^2/r^6$ one has $\nabla g = -4y_3 e_3/r^6 + 12y_3^2 y/r^8$, so $\nabla \times (gy) = \nabla g \times y = -4y_3(e_3 \times y)/r^6$. With $f = 4y_3/r^4$ one has $\nabla f = 4e_3/r^4 - 16y_3 y/r^6$, so $\nabla \times (fe_3) = \nabla f \times e_3 = -16y_3(y \times e_3)/r^6 = 16y_3(e_3 \times y)/r^6$. The total is $12y_3(e_3 \times y)/r^6$. Since $W \cdot \nabla W = \nabla \frac{|W|^2}{2} - W \times \omega$, it follows that $\nabla \times (W \cdot \nabla W) \neq 0$, i.e. $W \cdot \nabla W$ is not a gradient, i.e. $\mathbb{P}(W \cdot \nabla W) \neq 0$. $\square$

Consistency with Theorem 9. The curl of $\mathbb{P}(\tilde{h} \cdot \nabla \tilde{h})$ has symbol $i\zeta \times (iP_\zeta \tau \zeta) = -\zeta \times (\tau \zeta)$, using $\zeta \times P_\zeta A = \zeta \times A$. With $\|v_0\| = 1$, Theorem 9 makes this $-\frac{3\pi^3}{8}\zeta_3(\zeta \times e_3)/|\zeta|$. Writing its a -component as $-\frac{3\pi^3}{8}\epsilon_{ab3}\zeta_b\zeta_3/|\zeta|$ and using $|\zeta|^{-1} = c_{3,1}r^{-2} = 4\pi r^{-2}$ together with $\widetilde{\zeta_b\zeta_3 G} = -\partial_b\partial_3\tilde{G}$,

$$\left[\frac{\zeta_b\zeta_3}{|\zeta|}\right] = -4\pi\partial_b\partial_3r^{-2} = \frac{8\pi\delta_{b3}}{r^4} - \frac{32\pi y_3 y_b}{r^6};$$

since $\epsilon_{ab3}\delta_{b3} = 0$, only the second term survives, leaving $-32\pi y_3(y \times e_3)_a/r^6 = +32\pi y_3(e_3 \times y)_a/r^6$. Multiplying by $-\frac{3\pi^3}{8}$ gives $-12\pi^4 y_3(e_3 \times y)/r^6$. On the other hand $\tilde{h} = \pi^2 W$ gives $\tilde{h} \cdot \nabla \tilde{h} = \pi^4 W \cdot \nabla W$ and hence $\nabla \times (\tilde{h} \cdot \nabla \tilde{h}) = -\pi^4 \nabla \times (W \times \omega) = -12\pi^4 y_3(e_3 \times y)/r^6$. The two derivations agree exactly, constants included.

Transfer to the cutoff family

Lemma 13 (the cutoff perturbation is bounded near $\zeta = 0$). For all $|\zeta| \leq \frac{1}{8}$ and all i, j ,

$$|T_{ij}(\zeta) - \tau_{ij}(\zeta)| \leq C_1 := 64\pi\|v_0\|^2,$$

uniformly over all admissible χ .

Proof. Since $F = \chi(|\cdot|)h$,

$$T_{ij}(\zeta) - \tau_{ij}(\zeta) = \int_{\mathbb{R}^3} h_i(\eta) h_j(\zeta - \eta) [\chi(|\eta|)\chi(|\zeta - \eta|) - 1] d\eta.$$

The bracket vanishes whenever $|\eta| \leq \frac{1}{2}$ and $|\zeta - \eta| \leq \frac{1}{2}$, because $\chi \equiv 1$ on $[0, \frac{1}{2}]$. For $|\zeta| \leq \frac{1}{8}$, the condition $|\eta| \leq \frac{1}{4}$ implies $|\zeta - \eta| \leq \frac{1}{8} + \frac{1}{4} = \frac{3}{8} < \frac{1}{2}$; hence the integrand is supported in $\{|\eta| \geq \frac{1}{4}\}$. There $|\zeta - \eta| \geq |\eta| - \frac{1}{8} \geq |\eta| - \frac{|\eta|}{2} = \frac{|\eta|}{2}$. With $|h| \leq \|v_0\| \cdot |\cdot|^{-2}$ and $|\chi\chi - 1| \leq 1$,

$$|T_{ij} - \tau_{ij}| \leq \|v_0\|^2 \int_{|\eta| \geq 1/4} \frac{d\eta}{|\eta|^2 |\zeta - \eta|^2} \leq 4\|v_0\|^2 \int_{|\eta| \geq 1/4} \frac{d\eta}{|\eta|^4} = 4\|v_0\|^2 \cdot 4\pi \int_{1/4}^\infty \frac{dr}{r^2} = 64\pi\|v_0\|^2,$$

which is C_1 . $\square$

Theorem 10 (non-degeneracy). *For every $v_0 \in \mathbb{R}^3 \setminus \{0\}$ and every admissible χ ,*

$$\mathbb{P}_{\mathbb{R}^3}(V \cdot \nabla V) \neq 0.$$

Explicitly, with $v_0 = \|v_0\|e_3$, one has $\zeta \times (T(\zeta)\zeta) \neq 0$ — hence $P_\zeta M(\zeta) \neq 0$ — at every point of the open set of positive measure

$$\mathcal{U} := \left\{ \zeta \in \mathbb{R}^3 : 0 < |\zeta| < \frac{\pi^2}{2048}, |\zeta_3| \geq \frac{|\zeta|}{2}, |\zeta \times e_3| \geq \frac{|\zeta|}{2} \right\}.$$

Proof. Let $\zeta \in \mathcal{U}$. By Theorem 9,

$$|\zeta \times (\tau(\zeta)\zeta)| = \frac{3\pi^3}{8} \|v_0\|^2 \frac{|\zeta_3| |\zeta \times e_3|}{|\zeta|} \geq \frac{3\pi^3}{8} \|v_0\|^2 \cdot \frac{|\zeta|}{4} = \frac{3\pi^3}{32} \|v_0\|^2 |\zeta|.$$

Since $\frac{\pi^2}{2048} \approx 4.8 \times 10^{-3} < \frac{1}{8}$, Lemma 13 applies, and the Frobenius norm of $T(\zeta) - \tau(\zeta)$ is at most $3C_1$ (nine entries each at most C_1). Hence

$$|\zeta \times ((T(\zeta) - \tau(\zeta))\zeta)| \leq |\zeta| \cdot 3C_1 |\zeta| = 192\pi \|v_0\|^2 |\zeta|^2,$$

and therefore

$$|\zeta \times (T(\zeta)\zeta)| \geq \|v_0\|^2 |\zeta| \left(\frac{3\pi^3}{32} - 192\pi |\zeta| \right) > 0 \quad \text{whenever } |\zeta| < \frac{3\pi^3}{32 \cdot 192\pi} = \frac{3\pi^2}{6144} = \frac{\pi^2}{2048},$$

which holds on $\mathcal{U}$. The set $\mathcal{U}$ is open and nonempty (it contains, e.g., all small ζ in the direction $(1, 0, 1)/\sqrt{2}$), hence of positive measure. By Lemma 10, $P_\zeta M \neq 0$ on a set of positive measure, so $\mathbb{P}(V \cdot \nabla V) \neq 0$ in $L^2(\mathbb{R}^3)$. $\square$

Remark 12 (uniformity, and where the obstruction lives). *The proof of Theorem 10 is uniform in χ and in v_0 : the factor $\|v_0\|^2$ cancels between the main term and the perturbation, and the only properties of χ used are $0 \leq \chi \leq 1$ and $\chi \equiv 1$ on $[0, \frac{1}{2}]$ — no smoothness of χ enters §B.8 at all. Neither a specific χ nor any numerically evaluated convolution integral is required: the obstruction is carried entirely by the χ -independent leading singularity of $F * F$ at $\zeta = 0$, which is the same for every admissible cutoff. Moreover Theorem 9 shows the obstruction degenerates exactly on the two directions $\zeta \parallel v_0$ and $\zeta \perp v_0$, so the two cone conditions in $\mathcal{U}$ are essentially optimal.*

*This uniformity does **not** propagate to the constants of Theorem 11. It is the qualitative conclusion $\mathbb{P}(V \cdot \nabla V) \neq 0$ that holds uniformly in (χ, v_0) , and that is what licenses the quantifier “for every admissible χ and every $v_0 \neq 0$ ” there. The quantitative constants c_0 and N_* are extracted only after a test field Ψ has been chosen for the particular profile $V = V^{\chi, v_0}$ (Lemma 11), and they depend on (χ, v_0) ; no (χ, v_0) -independent value of either is claimed, and neither is effective (Remark 13).*

B.9 Proof of Theorem 2

Theorem 11 (capacity bound; restatement of Theorem 2). *Let $v_0 \in \mathbb{R}^3 \setminus \{0\}$, let χ be admissible (Definition 2), and let $u_N = u_N^{\chi, v_0}$ be the family (2). Then:*

- (1) u_N is a real, zero-mean, exactly divergence-free trigonometric polynomial banded at $|k| \leq N$, and the exact laws of Theorem 4 hold;

$$(2) \quad \frac{N}{44544} \leq N_0^2(u_N) \leq \frac{64}{3}N \text{ for every } N \geq 32;$$

(3) there exist $c_0 = c_0(\chi, v_0) > 0$ and $N_* = N_*(\chi, v_0) < \infty$ such that

$$\|\mathbb{P}(u_N \cdot \nabla u_N)\|_{L^2(\mathbb{T}^3)}^2 \geq c_0 N^3 \quad \text{for all } N \geq N_*.$$

Proof. (1) is Lemma 2 together with Theorem 4. (2) is Proposition 1.

(3) By Theorem 10, $\mathbb{P}_{\mathbb{R}^3}(V \cdot \nabla V) \neq 0$, where V is the profile of Definition 4 attached to the given (χ, v_0) . By Lemma 11 there is a real $\Psi \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$, divergence-free, supported in $\{|y| \leq R\}$ for some $R \geq 1$, with $I_\Psi \neq 0$. Fix such a Ψ . Lemma 8 and Theorem 7 give the exact inner rescaling with $\varepsilon_N + \varepsilon'_N = O(N^{-1} \log N)$; Lemma 9 and Theorem 8 convert the pairing of $\mathbb{P}(u_N \cdot \nabla u_N)$ against the concentrated field ψ_N into $(2\pi)^{-3} N^{3/2} (I_\Psi + \mathcal{E}_N)$ with $\mathcal{E}_N \rightarrow 0$; and Proposition 2 then yields the stated bound with

$$c_0 = c_0(\chi, v_0) = \frac{|I_\Psi|^2}{2(2\pi)^3 \|\Psi\|_{L^2(\mathbb{R}^3)}^2}, \quad N_* = N_*(\chi, v_0) = \min \left\{ N_0 \geq \max(8R, 32) : |\mathcal{E}_N| \leq (1-2^{-1/2})|I_\Psi| \forall N \geq N_0 \right\}$$

Both depend on (χ, v_0) alone, through $V = V^{\chi, v_0}$ and through the test field Ψ selected for it. $\square$

Remark 13 (what the constants depend on, and that they are non-effective). *The constants c_0 and N_* produced above depend on the pair (χ, v_0) only, through the profile $V = V^{\chi, v_0}$ and through the test field Ψ selected in Lemma 11. They are **not effective**. The reason is localised precisely at Lemma 11: Ψ is obtained from the density of $C_{c,\sigma}^\infty(\mathbb{R}^3)$ in $L_\sigma^2(\mathbb{R}^3)$ (Remark 9), a pure existence argument, so neither Ψ , nor its support radius R , nor I_Ψ , nor $\|\Psi\|_{L^2}$, nor c_0 , nor N_* is exhibited. Every other constant in this appendix is explicit, namely those of Lemmas 5–7, Proposition 1, Theorems 4, 5, 6, 7, 8, 9, 10, Lemmas 8, 9, 10, 12, 13, and Proposition 2 as a function of (Ψ, R, I_Ψ) .*

An effective variant is available in principle — take Ψ with $\hat{\Psi}$ a smooth bump supported in the explicit set $\mathcal{U}$ of Theorem 10, mollified to compact support in y — but it is expensive: non-vanishing is certified only for $|\zeta| < \pi^2/2048 \approx 4.8 \times 10^{-3}$, so such a Ψ has Fourier support at that scale, hence spatial radius $R \gtrsim 10^3$, hence $N_ \geq 8R \gtrsim 10^4$, with a correspondingly small c_0 . This is the sole source of non-effectivity in the paper.*

Remark 14 (the exponent 3 is sharp, and the sharp cutoff is not covered). *The exponent cannot be improved: by Theorem 1(d) and Theorem 4, $\|\mathbb{P}(u_N \cdot \nabla u_N)\|_2^2 \leq \|u_N\|_\infty^2 H_1(u_N)$, and both $\|u_N\|_\infty$ and $H_1(u_N)$ are $O(N)$ by Proposition 1 and $\|u_N\|_\infty \leq \sum_{k \neq 0} |\hat{u}_N(k)| \leq \|v_0\| \Sigma_N^\chi \leq \|v_0\| S_N \leq 128 \|v_0\| N$; so the left-hand side is $O(N^3)$.*

Smoothness of χ is used in exactly one place, namely Theorem 7, through $|\nabla F| \lesssim c_\chi |\xi|^{-3}$. For the sharply truncated family $\chi = \mathbf{1}_{[0,1]}$ this bound fails, F jumping across $|\xi| = 1$; the Riemann-sum defect then carries a lattice-point (Gauss-circle) term, and V loses the rapidly decaying tail correction of Theorem 6, acquiring instead an oscillatory $|y|^{-2}$ tail. Nothing in §B.7 or §B.8 would change, both being insensitive to χ ; the sharp-cutoff case is thus a technical gap in the inner-rescaling step, not a structural one. It is left open because no statement in this paper requires it (see Remark 2).